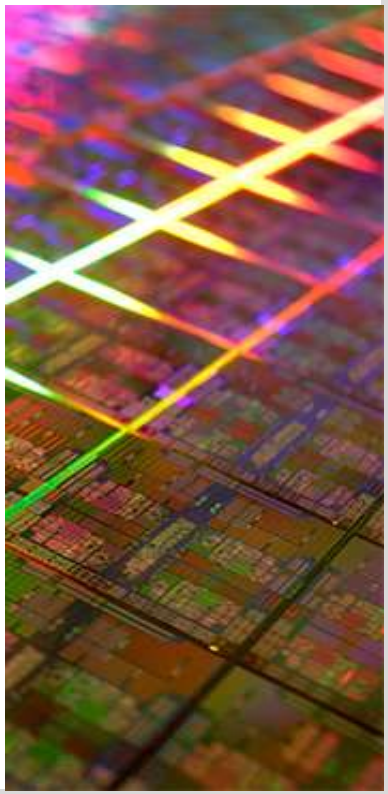
Specialty Chemicals and Gases for Semiconductor Devices Fabrication

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Multiple International Awards

23+ Years
Semiconductor Experience

Expertise
CMOS | 3D FinFETs
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27 Patents
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Outline

1 FAB INFRASTRUCTURE & UTILITIES



- Semiconductor fab and CUB infrastructure
- Cleanroom air system
- Cooling systems
- Major utility systems overview

2 WATER & UPW



- Types of water: Raw, Soft, RO, DI, UPW
- Water quality specifications
- Water treatment technologies
- Gas–UPW applications

3 GAS FUNDAMENTALS



- Functional classification of gases
- Bulk Gas vs. Specialty Gas
- Gas purity and contamination control
- Gas properties and flow units

4 BULK GASES IN SEMICONDUCTOR MANUFACTURING



- N_2, O_2, H_2, Ar, He, CDA
- CO_2 and rare gases: Ne, Kr, Xe
- Key functions
- Process applications

5 SPECIALTY GASES & PRECURSORS



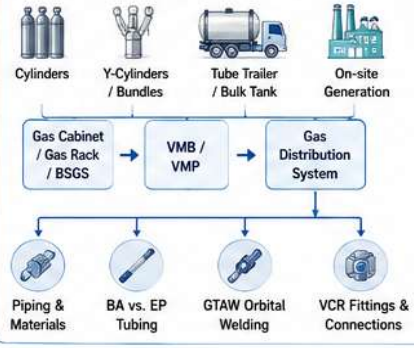
- Deposition precursors
- Reactive / plasma gases
- Etching gases
- Dopant gases
- Chamber-cleaning gases

6 GASES ACROSS SEMICONDUCTOR PROCESSES



- Different process
- Different gas chemistry
- Different function

7 GAS SUPPLY, STORAGE & DELIVERY SYSTEMS



8 SAFETY, EXHAUST & ABATEMENT SYSTEMS



THE JOURNEY OF A SEMICONDUCTOR GAS



OUR GOAL

Deliver the right utilities and environment to enable world-class semiconductor manufacturing.

Semiconductor Wafer Fab

Billions of dollars in CapEx involving millions of workhours within a short period of time

A semiconductor fab can take:

- >$20 billion overall CapEx investment
- $4-6 billion facility investment
- 30-40 million workhours

Month: 1[redacted]



- 580,000 m^3 of concrete:



3x
King Power MahaNakhon Tower
~60 months construction

- 83,000 tons of steel reinforcement:



1x
Golden Gate Bridge
~52 months construction

- >9,000 km of cabling:



Flight from Bangkok to Paris



BKK CDG

Fab Design & Construction (2-4 years) (Before production starts)

Planning

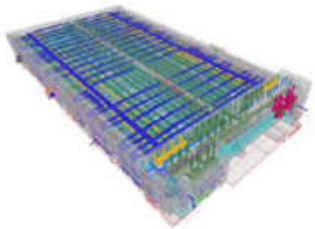


- Identify opportunities available, on-site infrastructure & natural resources
- Green incentives

5 - 7 months

Starts Planning

Design



- Fab requirements
- Detailed engineering
- System capacity sizing
- Permitting

6 - 20 months

Starts Design

Construction



- Building information modeling
- Off-site manufacturing
- Sourcing localization
- Green procurement

13 - 24 months

Starts Construction

Commissioning, Testing and Tool Hook Up

Starts Production

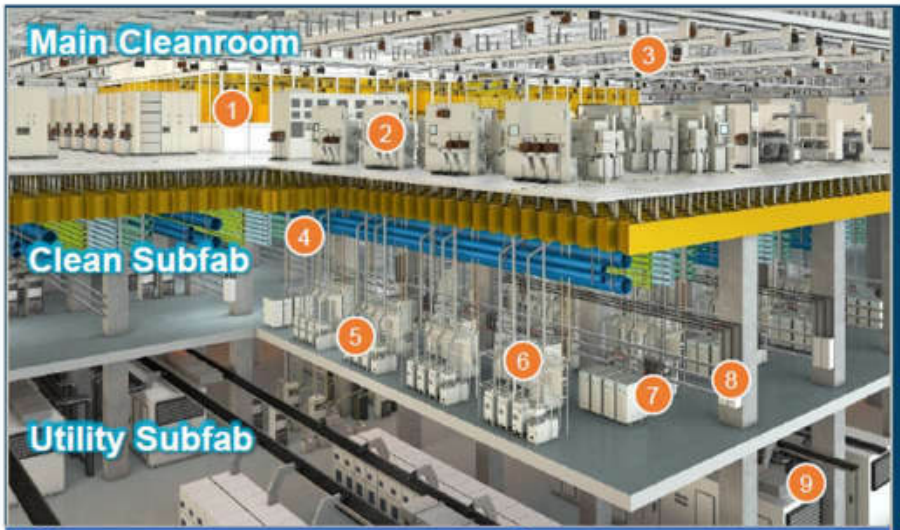
Production



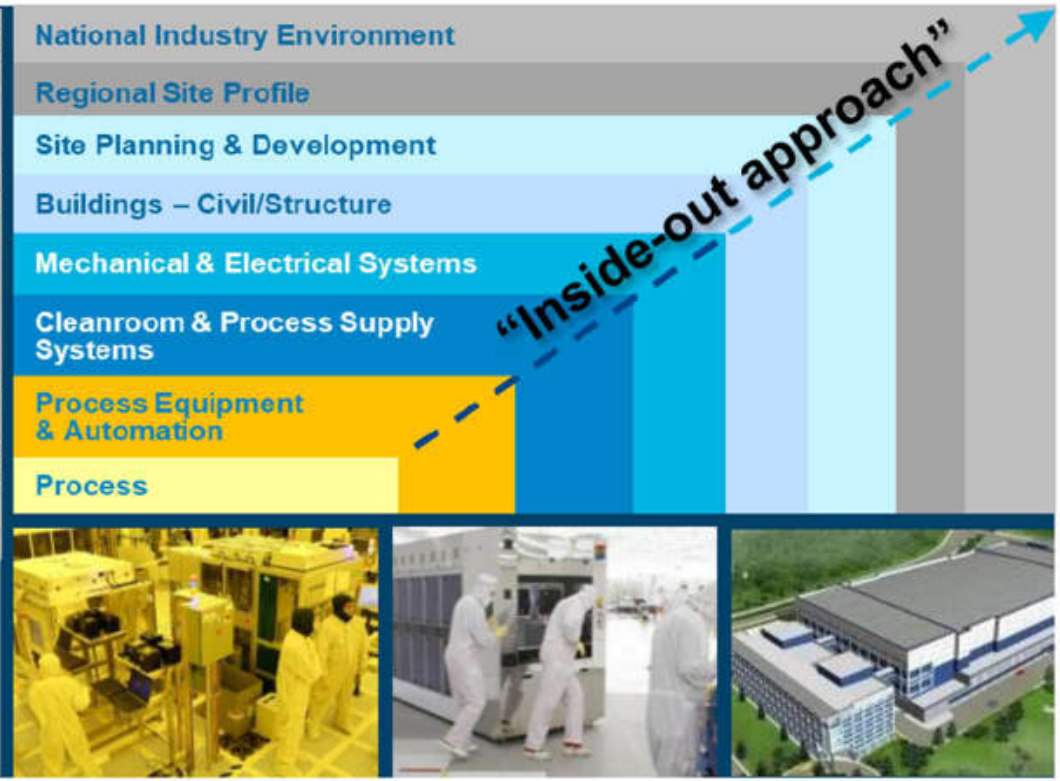
2 - 4 Years

Integrating All Aspects of Fab Facilities

Function, Operation & Shape of the Facility are driven by the Process !



1. Cleanroom
2. Process Equipment
3. Manufacturing Automation System
4. Mechanical System Distribution
5. Critical Subsystems
6. Tool Install
7. Process System Distribution
8. Electrical Distribution
9. Facility Plant Equipment



Wafer Fab Energy & Water Consumption

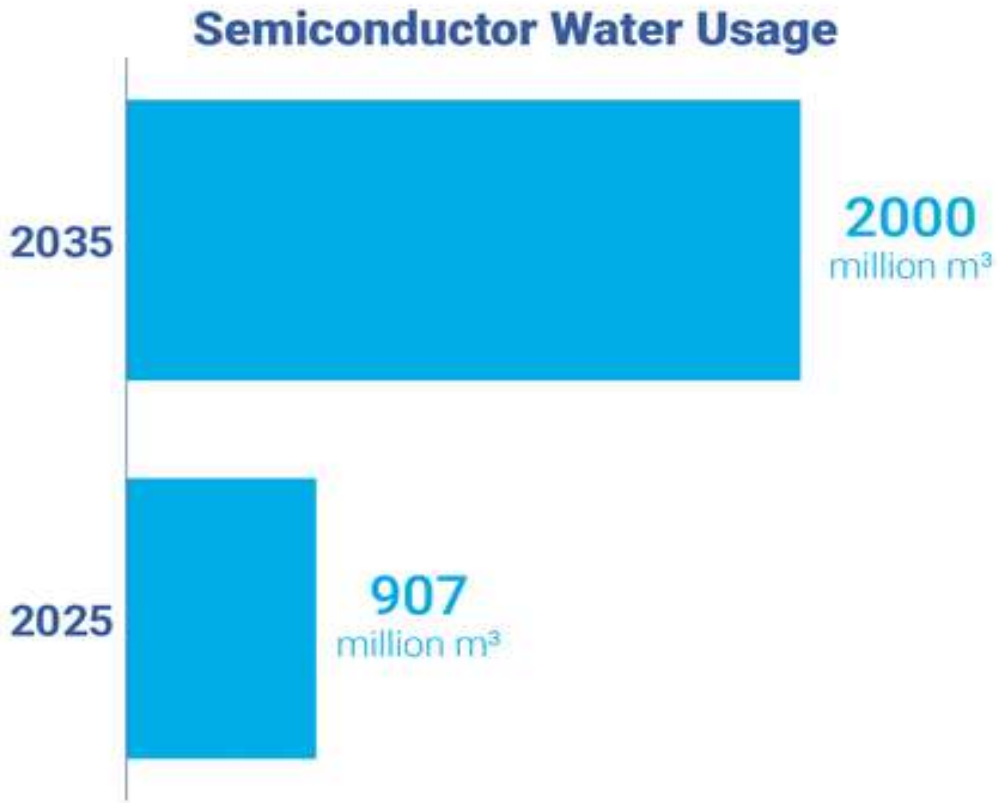
IDTechEx Research

Semiconductor manufacturing energy usage forecast to reach

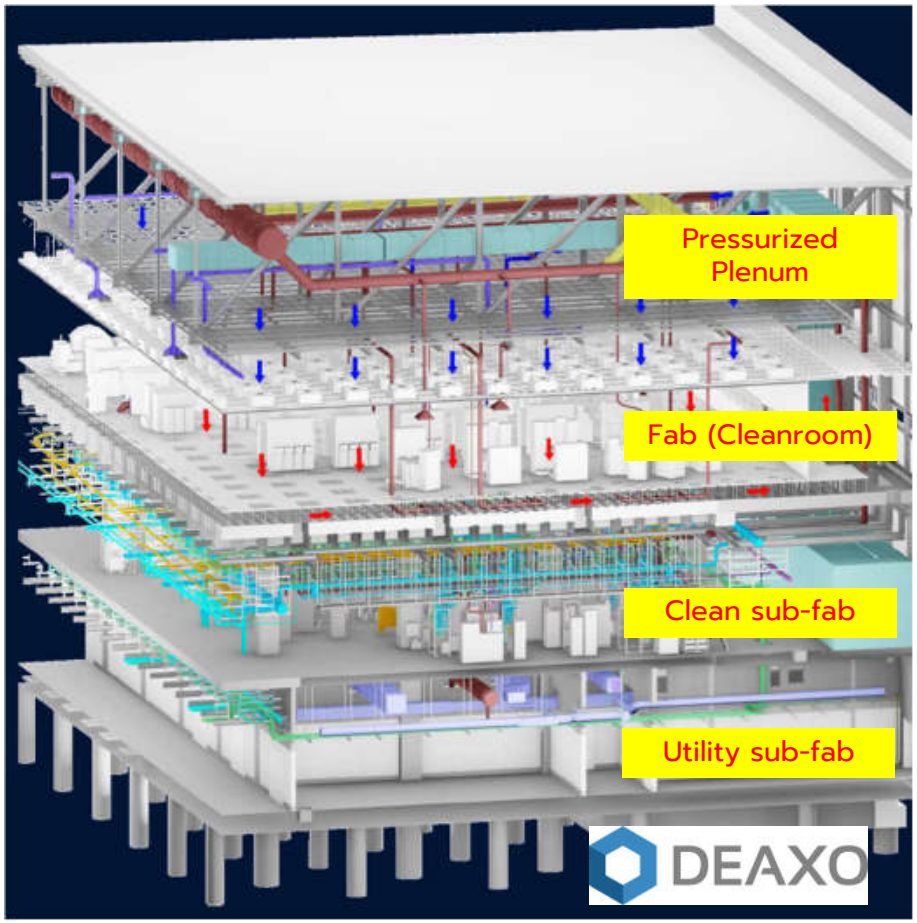
736 billion kWh

by **2035** at a **CAGR** of

12.0%



Central Utility Building (CUB) of Wafer Fab



























Major Utility Systems — CUB & Cleanroom Infrastructure

CUB = Generate / Condition / Treat → Cleanroom = Distribute / Control → Process Tool = Consume → Exhaust & Waste = Collect / Treat / Discharge

1. Power	MV/LV power, transformer, emergency generator, UPS
2. Cleanroom Environment	HVAC, MAU, AHU/RCU, FFU, HEPA/ULPA
3. Cooling	Chilled Water, Cooling Water, PCW
4. Water	Raw Water, Soft Water, RO, DI/UPW
5. Gases	CDA, N_2, O_2, Ar, H_2, He, specialty gases
6. Chemicals	Bulk Chemical Supply / CDS
7. Exhaust & Abatement	General, acid, solvent and process exhaust; POU scrubber
8. Waste	WWTP, acid/alkali, HF, solvent and chemical waste
9. Safety	Fire protection, GDS, emergency systems
10. Control & Monitoring	FMCS/BMS, utility monitoring, alarms

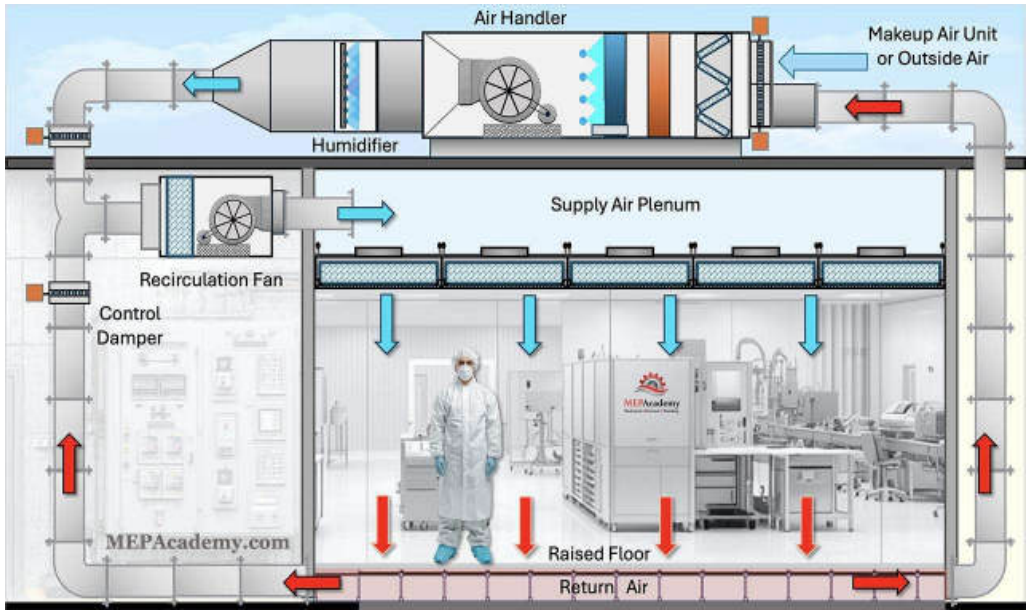
Cleanroom Air Supply, Recirculation, and Exhaust Units

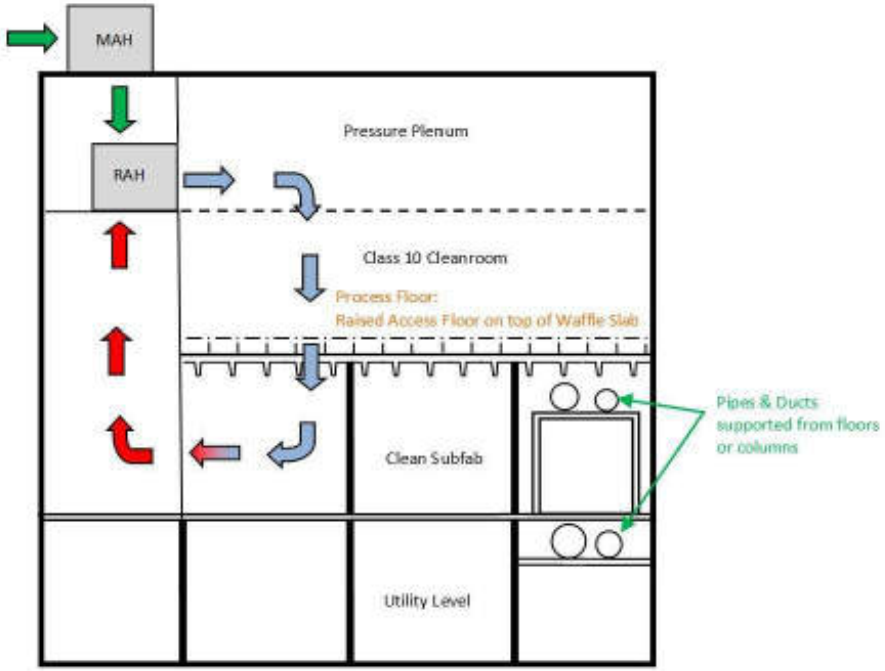
Wafer Fab Air System Units

Function	Main Equipment	What It Controls
1. Fresh / Make-up Air	OAC / MAU	Fresh air, humidity, dew point, pressurization
2. Central Air Conditioning	HVAC / AHU	Temperature, RH, airflow
3. Recirculation	RCU / Recirculation System	Large-volume continuous cleanroom airflow
4. Local Temperature Control	FCU / cooling coil	Local sensible heat load
5. Final Air Filtration	FFU + HEPA/ULPA	Airborne particles
6. Air Distribution	Supply plenum / ducts	Uniform air supply
7. Air Return	Return grille/floor + return shaft	Returns air for recirculation
8. Contaminant Removal	GEX / AEX / SEX / PEX	Heat, chemical vapor, particles and hazardous gases
9. Exhaust Treatment	POU / scrubber	Hazardous exhaust treatment
10. Final Discharge	Exhaust fan / stack	Safe atmospheric discharge

Cleanroom Air Recirculation

Most cleanroom air is not thrown away. A very large volume is continuously recirculated through FFU + HEPA/ULPA filters, while a smaller amount of conditioned fresh air from the OAC/MAU replaces the air removed by process and general exhaust.





Cleanroom Air System Unit

OAC	Outdoor Air Conditioner / Outdoor Air Conditioning Unit	Pre-conditions outside air before entering the cleanroom HVAC system	Outdoor air → conditioned fresh air	Utility area / mechanical floor	Controls temperature and especially humidity/dew point of incoming air
HVAC	Heating, Ventilation & Air Conditioning	Overall system controlling cleanroom environmental conditions	Integrates outdoor + recirculated air systems	CUB + mechanical floor + cleanroom	Controls temperature, RH, pressure, airflow and cleanliness
MAU	Make-Up Air Unit	Supplies conditioned outdoor air to replace air removed by exhaust	Outdoor air → MAU → cleanroom/recirculation system	Mechanical floor / utility building	Maintains air balance and positive cleanroom pressure
AHU	Air Handling Unit	Conditions and circulates air	Return air + sometimes fresh air	Mechanical floor / fan deck	Temperature/RH control, filtration and circulation
FCU	Fan Coil Unit	Provides localized sensible cooling/heating	Room/return air → coil → room	Cleanroom support area, office, utility room	Local temperature control; generally not primary cleanroom filtration
RCU	Recirculation Air Unit	Continuously recirculates large volumes of cleanroom air	Return air → RCU → filters/cleanroom	Sub-fab / mechanical floor	Maintains high cleanroom circulation rate and temperature stability
FFU	Fan Filter Unit	Provides filtered air directly into cleanroom	Plenum air → fan → HEPA/ULPA → cleanroom	Cleanroom ceiling	Main particle-control terminal unit
HEPA Filter	High-Efficiency Particulate Air Filter	Removes airborne particles	Supply/recirculation air	AHU/FFU/ceiling	Typically ≥99.97% removal at specified particle size
ULPA Filter	Ultra-Low Penetration Air Filter	Higher-efficiency particle filtration	Supply/recirculation air	FFU / cleanroom ceiling	Used for very stringent semiconductor cleanroom applications
Supply Air Plenum	—	Distributes conditioned air above cleanroom	AHU/RCU → plenum → FFU	Above cleanroom ceiling	Provides uniform air distribution to FFUs

Cleanroom Air Return & Exhaust Units

Return Air Unit / Path	RA / Return Air	Collects air leaving cleanroom and returns it to recirculation system	Cleanroom → return path → RCU/AHU	Raised floor / wall / return shaft	Completes cleanroom recirculation loop
Return Air Shaft	RAS	Vertical path for return air	Cleanroom/sub-fab → recirculation system	Cleanroom perimeter	Provides low-resistance return airflow
Recirculation System	RAS / RCU System	Reuses most cleanroom air instead of exhausting it	Cleanroom → return → cooling → FFU → cleanroom	Cleanroom + sub-fab + fan deck	Maintains cleanliness with very high airflow while reducing fresh-air conditioning load
General Exhaust	GEX / GE	Removes non-hazardous contaminated/heat-laden air	Room/tool → exhaust → atmosphere	Cleanroom/sub-fab → roof	Removes heat, odor and general contaminants
Acid Exhaust	AEX	Removes corrosive acid fumes	Wet bench/tool → duct → scrubber	Wet process areas	Handles HF, HCl, HNO_3, H_2SO_4, etc.
Solvent Exhaust	SEX / VEX	Removes solvent/VOC vapors	Tool/bench → duct → treatment	Lithography / solvent areas	Controls IPA, acetone, PGMEA and other VOCs
Process Exhaust	PEX	Removes hazardous process gases/by-products	Tool → POU abatement → exhaust	Process tools/sub-fab	Handles etch/deposition/process effluent
Scrubber / Abatement	POU / Central Scrubber	Treats hazardous exhaust before discharge	Contaminated exhaust → treatment → stack	Sub-fab / utility area	Removes acid, toxic, reactive gases and particles
Exhaust Stack	—	Final discharge of treated air	Exhaust system → atmosphere	Roof / external utility area	Safe dispersion of treated exhaust

Six major functional families of gases

Functional Family	Key Question	Examples
1. Material Source	What material do I want to add?	SiH_4, WF_6, $TiCl_4$, TMA
2. Reaction	What chemical reaction do I need?	O_2, NH_3, H_2, O_3
3. Material Removal	What material do I want to remove?	CF_4, SF_6, Cl_2, HBr
4. Doping	What impurity do I want to introduce?	PH_3, AsH_3, B_2H_6, BF_3
5. Process Support	How do I transport, purge, dilute, or protect?	N_2, Ar, He
6. Equipment Support	How do I enable plasma, cooling, cleaning, etc.?	Ar, He, NF_3, F_2

Cooling Liquids & Cooling Chemicals Used in Semiconductor Wafer Fabs

Cooling systems in a wafer fab

Facility cooling	Cooling Tower Water	Central Utility Building	Final heat rejection
Cleanroom / building cooling	Chilled Water	HVAC / MAU / AHU	Cleanroom temperature & humidity control
Process equipment cooling	PCW / DI cooling water	Process tools	Remove equipment/process heat
Precision thermal control	Glycol / fluorinated specialty fluids	Local tool chillers	Precise and/or sub-ambient temperature control

Cooling Liquids Used in Semiconductor Wafer Fabs

Cooling Tower Water (CTW)	Treated water + corrosion/scale/ biocide chemicals	~25–35°C	Reject facility heat to atmosphere	Cooling towers, condenser side of chillers	Scale, corrosion, biological growth
Chilled Water (CHW)	Treated closed-loop water	~5–15°C	General fab cooling / HVAC	AHU, MAU, cleanroom HVAC, facility equipment	Corrosion and microbial control
Process Cooling Water (PCW)	High-quality treated water, closed loop	~15–25°C	Remove heat from process equipment	Etcher, CVD/PVD, implanter, furnace, CMP, lithography, vacuum pumps	Temperature stability, pressure, cleanliness
High-Purity PCW	DI/RO-quality water in controlled loop	~15–25°C	Cooling where contamination/ corrosion is critical	RF generators, lasers, sensitive process equipment	Low conductivity; material compatibility
DI Water Cooling	Deionized water	~15–25°C	High-purity cooling	RF power supplies, lasers, magnets, selected tool components	Very low conductivity initially; corrosion/material compatibility

Cooling Chemicals Used in Semiconductor Wafer Fabs

Water–Ethylene Glycol (EG)	Water + ethylene glycol, often ~20–50%	~-20 to +100°C	Low-temperature cooling / freeze protection	Chillers, outdoor piping, sub-fab cooling	Toxicity, viscosity, heat-transfer penalty
Water–Propylene Glycol (PG)	Water + propylene glycol	~-15 to +100°C	Similar to EG with lower toxicity	Chillers / facility cooling	Higher viscosity than water
Fluorinated Heat-Transfer Fluid	Engineered fluorinated fluid	~-100 to +150°C, product-dependent	Precise wide-range thermal control	Etch/CVD chamber temperature control, wafer chuck/chiller	Expensive; environmental/GWP/PFAS considerations depend on chemistry
Hydrofluoroether / Engineered Fluids	HFE-type specialty fluids	Product-dependent	Electrical-safe precision cooling	Semiconductor chillers, electronics, thermal-control systems	Environmental/regulatory availability increasingly important
Silicone Oil	Polydimethylsiloxane-based fluid	~-50 to +200°C+	Wide-temperature heat transfer	Thermal baths, specialized equipment	High-temp stability but lower heat capacity than water
Mineral / Synthetic Oil	Hydrocarbon-based heat-transfer oil	~-20 to +300°C	High-temperature thermal control	Specialized process equipment, heaters/chillers	Flammability, leaks, contamination
Liquid N_2 (LN_2)	Cryogenic nitrogen	-196°C	Cryogenic cooling	Cryopumps, analytical equipment, specialized processes	Cryogenic/O_2-deficiency hazard; not normal PCW
Liquid He	Cryogenic helium	-269°C	Ultra-low-temperature cooling	Specialized analytical/research equipment	Very expensive; uncommon in production fab cooling

Ultrapure Water (UPW) & Dissolved-Gas DIW in Semiconductor Processing

Ultra-pure water (UPW)

UPW and Deionized water (DIW) is required due to the free of contaminants (ppb)



Unacceptable contaminants

- Dissolved ions (MIC): Na^+, Cl^-, K
- Heavy metals: Fe, Cu, Al, Cr, W, Ti
- Organic materials: C, Bacteria, lubricants, vapors, detergents, solvents, and moisture
- Particles: Cosmetic, Human, gases, chemicals
- Silica (reliability of oxidation).

UPW ≠ DIW

UPW (resistivity: 18.3 MΩ.cm, dissolved solid: 0.0277 ppm) remove dissolved oxygen gas to prevent native-oxide formation

8,000 liters of UPW is used to fabricate each 200 mm wafer

F. Li, *et. al.*, J. Electrochemical Soc., 152, 8, pp. G669-G673 (2005).

DIW: Deionized water UPW: Ultrapure water ppb: part-per-billion

Semiconductor Fab Water Classification & Typical Specifications

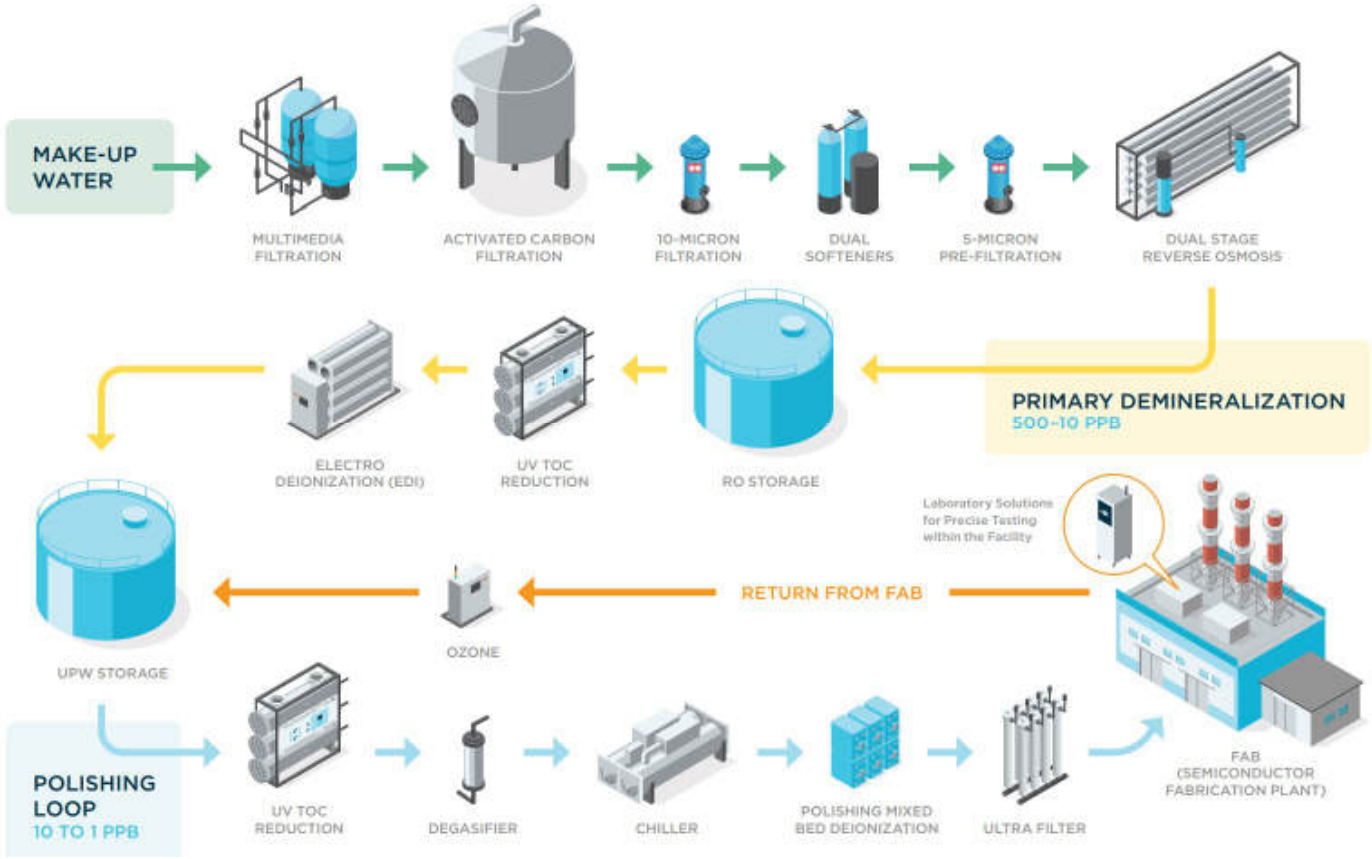
Level	Water	Main Contaminants Removed	Typical Resistivity	Semiconductor Application
1	Raw Water	—	Very low	Plant incoming water
2	Filtered Water	Particles, turbidity, chlorine	Very low	Pretreatment
3	Soft Water	Ca^{2+}, Mg^{2+} hardness	Very low	RO/boiler pretreatment
4	RO Water	95–99+% ions, organics, particles	~0.02–2 MΩ·cm	DI/UPW feed
5	DI Water	Dissolved ions	~1–18 MΩ·cm	General process cleaning/rinse
6	UPW	Ions + TOC + particles + bacteria + gases	≈18.2 MΩ·cm	Critical wafer processing

Resistivity and conductivity are reciprocal measures. The theoretical maximum resistivity of pure water at **25°C is approximately 18.2 MΩ·cm**, corresponding to conductivity of about **0.055 µS/cm**.

Semiconductor Fab Water Classification & Typical Specifications

Water Class	Typical Treatment / Source	Conductivity	Resistivity	TOC	Typical Key Characteristics	Typical Use in Fab
Raw Water	Municipal / well / surface water	100–1,000+ µS/cm	~0.001–0.01 MΩ·cm	~1–10+ mg/L	Contains Ca, Mg, Na, Cl, silica, organics, particles, microorganisms	Feed water to water-treatment plant; cooling tower makeup
Filtered Water	Multimedia / sand + activated carbon filtration	~100–800 µS/cm	~0.001–0.01 MΩ·cm	~0.5–5 mg/L	Suspended solids and chlorine reduced	Pretreatment / utility water
Soft Water	Ion-exchange softener removes Ca^{2+} and Mg^{2+}	Similar to raw water	Low	Similar to feed	Hardness typically <1–5 mg/L as $CaCO_3$; Na^+ remains	Boiler feed pretreatment, RO feed, utility systems
RO Water	Reverse Osmosis, typically 1-pass	~5–50 µS/cm	~0.02–0.2 MΩ·cm	~50–500 µg/L	Typically 95–99+% dissolved-ion rejection; low particles/organics	DI/UPW plant feed; some utility cleaning
Double-Pass RO	RO → RO	~0.5–10 µS/cm	~0.1–2 MΩ·cm	~20–200 µg/L	Much lower ions, silica and TOC	Preferred feed to semiconductor DI/UPW polishing
DI Water	Ion exchange / EDI / mixed bed	<1 µS/cm, often <0.1	~1–18 MΩ·cm	~10–100+ µg/L	Very low ionic contamination; organic/particle control depends on system	General process rinse, chemical preparation, equipment cleaning
High-Purity DI Water	RO + EDI/mixed bed + UV + filtration	~0.055–0.1 µS/cm	10–18.2 MΩ·cm	<10–50 µg/L	Very low ions and organics	Semiconductor process water / UPW make-up
Ultrapure Water (UPW)	RO + EDI/mixed bed + UV + degas + UF + final polishing	~0.055 µS/cm	≈18.2 MΩ·cm @ 25°C	typically <1–5 µg/L	Extremely low ions, organics, particles, bacteria and dissolved gases	Critical wafer rinsing, wet cleaning, chemical dilution, CMP/post-CMP
POU UPW	UPW + Point-of-Use final filtration/polishing	≈0.055 µS/cm	≈18.2 MΩ·cm	often ≤1–3 µg/L	Highest purity delivered immediately before process tool	Final wafer rinse / advanced semiconductor processes

Process flow diagram of an ultrapure water (UPW)



It takes roughly 1,400 to 1,600 gallons of municipal water to make 1,000 gallons of ultrapure water.

https://www.reddit.com/r/ChemicalEngineering/comments/1b1ixnz/found_this_process_flow_diagram_of_an_ultrapure/

Gas Used with UPW in Semiconductor Processing

Gas + UPW is not primarily about making UPW "purer"; it is about engineering the water chemistry for a specific process.

N_2-DIW	N_2	Deoxygenation / oxidation prevention	FE wet clean, metal processes
H_2-DIW	H_2	Reducing cleaning / oxidation suppression	Advanced FE cleaning
Ozonated DIW (DIO_3)	O_3	Organic removal / oxidative cleaning	FE wafer cleaning
Carbonated DIW	CO_2	Increase conductivity → ESD control	BE dicing / grinding / packaging

FE: Front-end process
BE: Back-end process

Gas Used with UPW in Semiconductor Processing

High-purity UPW is normally kept as pure as possible, while gases are typically introduced at a point-of-use (POU) module or process tool for a specific process.

N_2	N_2-saturated / N_2-bubbled UPW	Remove dissolved O_2 (DO); create low-oxygen water	Wet clean, rinse, metal-sensitive processes	FE + BE	Reduces oxidation/corrosion; improves surface control
H_2	H_2-dissolved UPW (H_2 water)	Create reducing environment; suppress oxidation	Advanced wafer cleaning, metal cleaning	Mainly FE	Reduces native oxide/oxidative damage and some contaminants
O_2	O_2-dissolved UPW (O_3-free oxygenated water)	Increase dissolved oxygen / promote controlled oxidation	Selected cleaning and surface-conditioning processes	Mainly FE	Controlled oxidation and surface chemistry
O_3	Ozone dissolved in UPW (DIO_3 / Ozonated DIW)	Strong oxidation of organic contamination	Wafer cleaning, photoresist residue removal, pre-clean	Mainly FE, some BE	Chemical reduction; effective organic removal
CO_2	CO_2-dissolved UPW (Carbonated DIW)	Increase conductivity and reduce static charge / ESD	Dicing, wafer thinning, grinding, CMP-related rinse	Mainly BE / Packaging	Prevents ESD and particle attraction
CO_2	Carbonated UPW	Lower/control pH	Surface cleaning/rinsing	FE + BE	Controls surface charge/zeta potential and particle behavior
$N_2 + H_2$	Forming-gas-related dissolved water concepts	Reducing environment	Specialized cleaning	FE / specialized	Oxidation suppression; surface conditioning
Ar	Ar-bubbled UPW	Degassing / displacement of dissolved gases	Specialized research/process cleaning	Limited	Chemically inert gas environment

FE: Front-end process
BE: Back-end process

Important Dissolved-Gas Technologies For advanced wafer cleaning

UPW is sometimes intentionally contacted with or dosed with gases to control dissolved oxygen (DO), dissolved CO_2, pH, oxidation/reduction behavior, surface charge, and electrostatic discharge (ESD).

Dissolved Gas in UPW/DIW	Common Name	Water Property Changed	Primary Benefit	Typical Application
O_3	DIO_3 / Ozonated DI Water	Strongly increases oxidation potential	Oxidizes/removes organic contamination	Organic clean, surface preparation
H_2	H_2-DIW / Hydrogen Water	Creates reducing environment	Particle removal and surface oxidation control	Advanced cleaning + megasonic
CO_2	Carbonated DI Water	Increases conductivity / lowers resistivity	Reduces electrostatic charging (ESD)	Spin rinse, single-wafer clean
O_2	O_2-DIW	Controls dissolved oxygen / oxidation potential	Controls surface oxidation chemistry	Specialized wafer cleaning
N_2	N_2-saturated / deaerated water	Reduces dissolved O_2	Suppresses unwanted oxidation	Final rinse / oxidation-sensitive surfaces

Why CO_2 is especially important in backend

UPW has extremely high resistivity:
UPW ≈ 18.2 MΩ·cm → extremely poor electrical conductivity
During **wafer dicing, grinding, high-pressure spraying, or spin rinsing**, flowing UPW can generate electrostatic charge. This can cause electrostatic discharge **(ESD) damage** to sensitive devices.

Adding a controlled amount of **CO_2**:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

increases ionic conductivity and therefore **reduces water resistivity**, allowing accumulated charge to dissipate.

Pure UPW → high resistivity → static charge risk
CO_2 + UPW → higher conductivity → lower static charge → ESD protection
This is why **carbonated DI water / CO_2-DIW is particularly common around wafer dicing and some backend wet processes**.

The need for ultrapure water in wafer fab





https://www.weforum.org/stories/manufacturing-and-value-chains/the-water-challenge-for-semiconductor-manufacturing-and-big-tech-what-needs-to-be-done/

Gas Used in Semiconductor Processing

Functional Classification of Gases in Semiconductor Front-End Manufacturing (1)

Gas Category	Definition	Main Purpose / Function	Typical Gases	Typical Processes
Carrier Gas	A gas used to transport another gas, vapor, or precursor into the process chamber	Deliver precursor uniformly; control concentration and flow	N_2, H_2, Ar, He	CVD, ALD, epitaxy, oxidation/diffusion
Precursor Gas	A gas or vapor containing the chemical species that forms the deposited film	Supply Si, metal, dielectric, dopant, etc. to the wafer surface	SiH_4, SiH_2Cl_2, TEOS vapor, WF_6, $TiCl_4$, TMA	CVD, PECVD, LPCVD, ALD, epitaxy
Reactive Gas (Reactant)	A gas that chemically reacts with a precursor or wafer material to form or modify a film	Oxidation, nitridation, reduction, film formation	O_2, O_3, NH_3, H_2, N_2O, H_2O vapor	CVD, ALD, oxidation, nitridation
Etchant Gas	A reactive gas used to remove material selectively from the wafer	Plasma or chemical etching of Si, oxide, nitride, metals, etc.	CF_4, CHF_3, C_4F_8, SF_6, Cl_2, BCl_3, HBr	RIE, ICP etch, dry etch
Purge Gas	An inert or non-reactive gas used to remove residual process gases, precursors, and by-products	Prevent unwanted reactions and cross-contamination	N_2, Ar	ALD, CVD, furnace, gas delivery systems
Dilution Gas	A gas used to reduce the concentration of a reactive, toxic, or flammable process gas	Improve process control and safety	N_2, Ar, H_2, He	CVD, PECVD, implant, gas mixtures

Functional Classification of Gases in Semiconductor Front-End Manufacturing (2)

Gas Category	Definition	Main Purpose / Function	Typical Gases / Examples	Typical Processes
Plasma Gas	A gas introduced primarily to generate and sustain plasma	Produce ions, electrons and radicals	Ar, He, N_2	Plasma etch, PECVD, PVD, plasma clean
Sputtering Gas	A gas ionized to bombard a solid target and eject atoms for deposition	Physical sputtering of target material	Ar; sometimes Kr/Xe	PVD / sputtering
Dopant Gas	A gas containing an element used to modify semiconductor electrical properties	Introduce n-type or p-type dopants	PH_3, AsH_3, B_2H_6, BF_3	Ion implantation, epitaxy, diffusion
Oxidizing Gas	A gas providing an oxidizing environment or oxygen-containing reactive species	Grow oxide or oxidize process materials	O_2, O_3, N_2O, H_2O vapor	Thermal oxidation, ALD, CVD, chamber clean
Reducing Gas	A gas used to create a reducing chemical environment	Remove/reduce oxides; assist metal formation	H_2, forming gas (H_2/N_2)	Annealing, epitaxy, CVD, surface treatment
Nitriding Gas	A nitrogen-containing reactive gas used to incorporate nitrogen into a material	Form nitride films or modify interfaces	NH_3, N_2, N_2/H_2 plasma	SiN deposition, ALD, nitridation

Functional Classification of Gases in Semiconductor Front-End Manufacturing (3)

Gas Category	Definition	Main Purpose / Function	Typical Gases / Examples	Typical Processes
Cleaning Gas	A gas used to remove deposits/residues from process chambers or wafer surfaces	Chamber cleaning and residue removal	NF_3, F_2, ClF_3, CF_4/O_2	CVD/PECVD chamber clean, remote plasma clean
Inert / Protective Gas	A chemically stable gas used to provide a non-reactive atmosphere	Prevent oxidation, contamination or unwanted reactions	N_2, Ar, He	Furnace, PVD, load lock, transfer chamber
Backside Gas	A gas introduced between the wafer backside and electrostatic chuck	Improve wafer-to-chuck thermal conduction	He	Plasma etch, PECVD, high-density plasma processes
Annealing / Ambient Gas	Gas used to establish the desired atmosphere during thermal treatment	Control oxidation, reduction, passivation and material properties	N_2, Ar, H_2, forming gas, O_2	Furnace anneal, RTA/RTP, forming-gas anneal
Laser Gas	Gas used as the active medium or supporting gas in lithography laser systems	Generate DUV laser radiation	Ne + F_2 + Ar/Kr/Xe mixtures	KrF/ArF excimer lithography

Six Major Functional Families of Gases

Functional Family	Key Question	Examples
1. Material Source	What material do I want to add?	SiH_4, WF_6, $TiCl_4$, TMA
2. Reaction	What chemical reaction do I need?	O_2, NH_3, H_2, O_3
3. Material Removal	What material do I want to remove?	CF_4, SF_6, Cl_2, HBr
4. Doping	What impurity do I want to introduce?	PH_3, AsH_3, B_2H_6, BF_3
5. Process Support	How do I transport, purge, dilute, or protect?	N_2, Ar, He
6. Equipment Support	How do I enable plasma, cooling, cleaning, etc.?	Ar, He, NF_3, F_2

Gas Used in Wafer Wet Cleaning & Wet Etching Process

Gases Used in Wafer Wet Cleaning & Wet Etching (1)

N_2	Purge / Drying / Carrier	DI/UPW, wet bench, IPA vapor	Rinse, drying, chemical dispense	Remove moisture; prevent contamination/oxidation	Most widely used support gas in wet cleaning
O_2	Oxidizing / Dissolved gas	DI/UPW, chemical solutions	Oxidative cleaning, surface conditioning	Oxidize Si surface and organic contaminants	Dissolved O_2 strongly affects Si surface chemistry
O_3	Strong oxidizing gas	Ozonated DI Water (DIO_3)	Organic clean, photoresist residue removal	Organics, hydrocarbons, metallic contamination assistance	Important alternative/reduction approach to conventional strong chemical cleans
H_2	Reducing / Dissolved gas	Hydrogenated DI Water (H_2-DIW)	Advanced wafer cleaning/rinsing	Particle removal, control oxidation potential	Used in advanced single-wafer cleaning; often combined with megasonic
CO_2	Conductivity / ESD control	CO_2-dissolved DI/UPW	Rinsing, spin cleaning, wafer drying	Increase water conductivity; suppress electrostatic charging	Particularly useful because UPW has extremely high resistivity

Gases Used in Wafer Wet Cleaning & Wet Etching (2)

NH_3	Alkaline reactive gas / chemical source	Dissolved in water → NH_4OH	SC-1 / APM cleaning	Particles, organics, some metals	NH_4OH + H_2O_2 + H_2O is conventional SC-1
HCl	Acid reactive gas / chemical source	Dissolved in water → HCl solution	SC-2 / HPM cleaning	Metallic contamination	HCl + H_2O_2 + H_2O is conventional SC-2
HF	Acidic etchant vapor / chemical source	HF aqueous solution / HF vapor	Native oxide removal, SiO_2 etching	SiO_2 / native oxide	Central chemistry for oxide removal; aqueous HF/DHF is a wet process
H_2O vapor / Steam	Oxidizing / Hydration	Water vapor / hot DI water	Surface treatment, rinse / specialized cleaning	Surface hydroxylation and residue removal	More specialized than conventional RCA cleaning
IPA Vapor	Drying vapor*	IPA + N_2	Marangoni drying / vapor drying	Water removal; reduce watermark formation	Very important after final DI rinse
N_2 + IPA Vapor	Carrier + drying system	IPA vapor	Marangoni / vapor drying	Particle/watermark control	N_2 commonly transports and controls IPA vapor
Air / CDA	Drying / Pneumatic support	Wet station	General drying / equipment operation	Moisture removal / equipment actuation	N_2 generally preferred for critical wafer surfaces

Gas Used in Thin Film Deposition Process

Semiconductor Film Deposition Methods (1)

Thermal Oxidation	Thermal Oxidation	Excellent Si/SiO_2 interface; high-quality dense oxide; excellent uniformity	High temperature; mainly limited to oxidation of Si; consumes Si substrate	SiO_2	Gate oxide, field oxide, pad oxide, sacrificial oxide
APCVD	Atmospheric Pressure Chemical Vapor Deposition	High deposition rate; simple reactor; high throughput	More particle/gas-phase reaction; poorer step coverage than low-pressure processes	SiO_2, doped oxide, Si	Dielectric/passivation films, legacy processes
LPCVD	Low-Pressure Chemical Vapor Deposition	Excellent uniformity; dense/high-quality films; good conformality; batch processing	High process temperature; relatively long cycle; thermal budget	Poly-Si, Si_3N_4, SiO_2, TEOS oxide	Gate/poly structures, spacers, hard masks, dielectric layers
PECVD	Plasma-Enhanced Chemical Vapor Deposition	Low deposition temperature; high rate; wide material range	Film quality/density generally below thermal/LPCVD; plasma damage possible; higher H content	SiO_2, SiN_x, SiON, a-Si:H, SiC:H	Passivation, ILD, hard mask, etch-stop, dielectric layers
HDP-CVD	High-Density Plasma Chemical Vapor Deposition	Excellent gap fill; simultaneous deposition and sputtering; dense film	Complex equipment; plasma damage; higher cost	Primarily SiO_2-based dielectric	STI/gap fill, interlayer dielectric
SACVD	Sub-Atmospheric Chemical Vapor Deposition	Good conformality and gap filling; relatively high rate	Chemistry/process complexity; less precise thickness control than ALD	O_3/TEOS SiO_2, doped oxide	Gap fill, ILD, dielectric deposition
RTCVD	Rapid Thermal Chemical Vapor Deposition	Short thermal cycle; reduced thermal budget; good process control	Lower throughput; complex temperature control	Si, SiGe, dielectric films	Epitaxy, advanced CMOS structures
Epitaxy / Epi-CVD	Chemical Vapor Deposition Epitaxy	High-quality single-crystal films; precise composition/doping; selective growth possible	High temperature; stringent surface cleanliness; complex chemistry	Si, SiGe, doped Si	Raised S/D, strain engineering, buried layers, advanced CMOS
MOCVD / MOVPE	Metal-Organic Chemical Vapor Deposition / Metal-Organic Vapor Phase Epitaxy	Excellent compound-semiconductor crystal quality; composition/doping control; scalable production	Expensive precursors; toxic/pyrophoric gases; high temperature; complex gas management	GaN, AlGaN, InGaN, GaAs, InP	LEDs, laser diodes, GaN power, RF, III-V devices

Semiconductor Film Deposition Methods (2)

MBE	Molecular Beam Epitaxy	Extremely precise layer/composition control; excellent interfaces; atomic-scale growth; in-situ monitoring	Very slow; expensive; UHV required; low throughput	GaAs, AlGaAs, InGaAs, InP, GaN, SiGe	Quantum wells, superlattices, research, lasers, high-performance III-V devices
ALD	Atomic Layer Deposition	Excellent conformality; Å-level thickness control; excellent 3D coverage	Very slow deposition rate; long cycle time; expensive precursors	Al_2O_3, HfO_2, ZrO_2, TiO_2, TiN, TaN	High-k gate dielectric, capacitor, barrier/liner, advanced 3D structures
PEALD	Plasma-Enhanced Atomic Layer Deposition	ALD-level conformality plus lower temperature; wider chemistry; improved reactivity	Plasma damage possible; equipment more complex; conformality may depend on radical transport	Al_2O_3, HfO_2, SiN_x, TiN, TaN	GAA/FinFET, high-k, spacers, liners, barriers
PVD – Sputtering	Physical Vapor Deposition – Sputtering	High-purity films; good metal deposition rate; mature; relatively simple; many target materials	Line-of-sight limitation; poor coverage in high-aspect-ratio structures; ion damage possible	Al, Cu, Ti, Ta, W, Ni, Co	Metal interconnect, contact, barrier, liner, seed layer
Reactive PVD	Reactive Physical Vapor Deposition / Reactive Sputtering	Easy formation of compound films; mature technology	Target poisoning; process stability issues; limited conformality	TiN, TaN, AlN, oxides	Diffusion barrier, electrode, hard mask
E-Beam Evaporation	Electron-Beam Physical Vapor Deposition	High-purity film; high melting-point materials possible; low substrate heating	Strong line-of-sight; poor step coverage; possible X-ray/electron effects	Au, Al, Ti, Cr, Ni, Pt	Metallization, lift-off, MEMS, sensors, R&D
Thermal Evaporation	Thermal Physical Vapor Deposition	Simple; relatively low cost; high-purity film; good for lift-off	Limited source materials; poor step coverage; line-of-sight	Al, Au, Ag and other low/moderate-melting metals	Metal electrodes, MEMS, sensors, lift-off
Metal CVD	Metal Chemical Vapor Deposition	Better conformality than PVD; good filling of small features	Complex/toxic precursors; contamination concerns; higher process cost	W, Co, Ru, TiN	Contact/via fill, liner, electrode, advanced interconnect
Electroplating / ECD	Electrochemical Deposition	Very high deposition rate; excellent Cu filling; low cost per wafer for thick films	Requires conductive seed layer; wet chemistry; contamination/control complexity	Cu, Ni, Co, Au	Cu interconnect, TSV, RDL, Cu pillar*

Thin-Film Deposition Methods Comparison

Technology	Film Quality	Conformality	Deposition Rate	Temperature	Thickness Control	Typical Strength
Oxidation	★★★★★	★★★★★	★★–★★★	High	★★★★	Highest-quality SiO_2
LPCVD	★★★★★	★★★★	★★★	High	★★★★	Dense, uniform films
PECVD	★★★	★★★	★★★★	Low	★★★	Low-temperature dielectric
HDP-CVD	★★★★	★★★★	★★★	Medium	★★★	Gap filling
Epitaxy	★★★★★	–	★★★	High	★★★★	Single-crystal Si/SiGe
MOCVD	★★★★★	★★★	★★★	High	★★★★	III-V epitaxy
MBE	★★★★★	★★★	★	Low–Medium	★★★★★	Atomic-level epitaxy
ALD	★★★★★	★★★★★	★	Low–Medium	★★★★★	3D conformality
PEALD	★★★★–★★★★★	★★★★★	★–★★	Low	★★★★★	Low-T ALD
PVD	★★★★	★★	★★★★	Low–Medium	★★★	Metal deposition
Evaporation	★★★★	★	★★★★	Low	★★★	Lift-off/metallization
Metal CVD	★★★★	★★★★	★★★	Medium	★★★★	Conformal metal
ECD	★★★★	★★★★*	★★★★★	Low	★★★	Thick Cu / feature fill

*ECD can provide excellent bottom-up filling when seed, additives, chemistry, and feature geometry are properly controlled

The easiest Way to Remember the Thin-Film Deposition Technologies

Thermal Growth	React wafer surface	Best Si/SiO_2 interface	SiO_2
CVD	Gas-phase precursor → surface reaction	High throughput + good film quality	SiO_2, SiN, Poly-Si, W
PECVD	Plasma activates CVD chemistry	Low temperature	SiO_2, SiN, SiON
Epitaxy	Grow crystal on crystal	Single-crystal film	Si, SiGe
MOCVD	Metal-organic precursor epitaxy	III-V production	GaN, GaAs, InP
MBE	Molecular/atomic beams in UHV	Ultimate epitaxial control	III-V, quantum structures
ALD	Sequential self-limiting reactions	Best conformality + atomic thickness control	High-k, oxide, nitride
PEALD	Plasma-assisted ALD	Low-T conformal deposition	High-k, SiN, TiN
PVD	Physical ejection from target	Metal deposition	Al, Cu, Ti, Ta
Evaporation	Vaporize solid source	Lift-off / simple metal stack	Au, Al, Ti, Cr
ECD	Electrochemical reduction	Thick metal / Cu filling	Cu, Ni, Au

Gases Used in Major Semiconductor Film Deposition Processes (1)

Deposition Process	Typical Films	Precursor / Source Gases	Reactant Gases	Carrier / Dilution Gas	Purge / Plasma / Other Gas	Key Feature
Thermal Oxidation / Oxidator	SiO_2	—	O_2, H_2O/steam	N_2	N_2	Thermal oxidation of Si; dry or wet oxidation
LPCVD	Poly-Si, SiO_2, Si_3N_4, TEOS oxide	SiH_4, SiH_2Cl_2 (DCS), TEOS vapor	NH_3, N_2O, O_2	N_2, H_2	N_2	High-T, low-pressure thermal deposition
APCVD	SiO_2, doped oxide	SiH_4, TEOS	O_2, O_3, PH_3, B_2H_6	N_2	N_2	Atmospheric-pressure CVD
PECVD	SiO_2, SiN_x, SiON, a-Si:H	SiH_4, TEOS	NH_3, N_2O, O_2, N_2	N_2, Ar, He	Ar, He, N_2	Plasma enables lower deposition temperature
HDP-CVD	SiO_2 / gap-fill oxide	SiH_4	O_2	—	Ar, He	Simultaneous plasma deposition + ion bombardment
SACVD	SiO_2 / gap fill	TEOS	O_3	N_2	N_2	Sub-atmospheric CVD; good conformality
Epitaxy – Si	Epi-Si, SiGe	SiH_4, DCS, $SiHCl_3$ (TCS)	H_2	H_2	HCl	High-quality crystalline film
Epitaxy – SiGe	SiGe	SiH_4/DCS + GeH_4	H_2	H_2	HCl; B_2H_6/PH_3 for doping	Strained-channel engineering
MOCVD	GaN, AlGaN, InGaN, GaAs	TMGa, TMAl, TMIn vapors	NH_3, AsH_3, PH_3	H_2, N_2	N_2/H_2	III-V compound semiconductor growth

Gases Used in Major Semiconductor Film Deposition Processes (2)

Deposition Process	Typical Films	Precursor / Source Gases	Reactant Gases	Carrier / Dilution Gas	Purge / Plasma / Other Gas	Key Feature
MBE	GaAs, AlGaAs, InGaAs, GaN	Usually solid elemental beams: Ga, Al, In etc.	As_2/As_4, activated N species	Usually none	N_2 for plasma source in PAMBE	UHV; molecular/atomic beam growth
Thermal ALD	Al_2O_3, HfO_2, ZrO_2, TiN, etc.	TMA, TEMAH/$HfCl_4$, $TiCl_4$, etc.	H_2O, O_3, NH_3	—	N_2 or Ar purge	Self-limiting surface reactions
PEALD	Al_2O_3, HfO_2, TiN, SiN_x, etc.	TMA, TEMAH, $TiCl_4$, BTBAS, etc.	O_2, H_2, N_2, NH_3	Ar/N_2	O_2, H_2, N_2, NH_3, Ar plasma	Plasma-enhanced ALD; lower temperature
PVD Sputtering	Al, Cu, Ti, Ta, W, TiN, TaN	Solid target	N_2/O_2 for reactive sputtering	—	Ar	Ar ions sputter atoms from target
Reactive PVD	TiN, TaN, AlN, oxides	Solid metal target	N_2 or O_2	—	Ar	Reactive gas forms compound film during sputtering
E-beam Evaporation	Al, Au, Ti, Ni, Cr etc.	Solid source	Usually none	—	Usually none	High-vacuum physical evaporation
Thermal Evaporation	Al, Au, Ag etc.	Solid source	Usually none	—	Usually none	Thermal evaporation of source material
Metal CVD	W, TiN, Co, Ru etc.	WF_6, $TiCl_4$, metal-organic precursors	H_2, NH_3, SiH_4	N_2/Ar	N_2/Ar	Conformal metal/barrier deposition
Tungsten CVD	W	WF_6	H_2, SiH_4	N_2/Ar	N_2	Contact/via fill
TiN CVD/ALD	TiN	$TiCl_4$ or metal-organic Ti precursor	NH_3	N_2/Ar	N_2/Ar	Barrier/electrode film
Selective Epitaxy	Si, SiGe	DCS/SiH_4 + GeH_4	H_2	H_2	HCl	Source/drain epi in advanced CMOS

Typical Gases by Film Material (1)

For process engineers, another useful way is to start with **"What film do we want to deposit?"**

SiO_2	Oxidation	Si substrate	O_2 / H_2O
SiO_2	PECVD	SiH_4 / TEOS	N_2O / O_2
SiO_2	LPCVD	TEOS / SiH_4	O_2/N_2O
Si_3N_4 / SiN_x	LPCVD	DCS	NH_3
SiN_x	PECVD	SiH_4	NH_3/N_2
Poly-Si	LPCVD	SiH_4	— / H_2
Epi-Si	Epitaxy	DCS/TCS/SiH_4	H_2
SiGe	Epitaxy	Si precursor + GeH_4	H_2
Al_2O_3	ALD/PEALD	TMA	$H_2O/O_3/O_2$ plasma
HfO_2	ALD/PEALD	$HfCl_4$/TEMAH-type precursor	$H_2O/O_3/O_2$ plasma
TiN	CVD/ALD	$TiCl_4$ / organometallic Ti precursor	$NH_3/N_2/H_2$ plasma, chemistry-dependent

Typical Gases by Film Material (2)

For process engineers, another useful way is to start with **"What film do we want to deposit?"**

W	CVD	WF_6	H_2/SiH_4
Ti	PVD	Ti target	Ar
TiN	Reactive PVD	Ti target	Ar + N_2
Ta	PVD	Ta target	Ar
TaN	Reactive PVD	Ta target	Ar + N_2
Cu	PVD	Cu target	Ar
Al	PVD	Al target	Ar
GaN	MOCVD	TMGa	NH_3 + H_2/N_2
AlGaN	MOCVD	TMAl + TMGa	NH_3 + H_2/N_2
InGaN	MOCVD	TMIn + TMGa	NH_3 + N_2/H_2, recipe-dependent
GaAs	MOCVD	TMGa	AsH_3
GaAs	MBE	Ga source	As_2/As_4

Gas Used in Thin Film Deposition Process

1. Silane (SiH_4)

for CVD/PECVD of silicon-containing films (>5N)

Poly-Si	LPCVD	Si precursor	— / H_2	$SiH_4 \rightarrow Si + 2H_2$	Gate, electrode, MEMS structures
Amorphous Si (a-Si:H)	PECVD	Si precursor	H_2	Plasma decomposition of SiH_4	Sacrificial/functional films, sensors
SiO_2	PECVD	Si precursor	N_2O / O_2	SiH_4 + oxidant → SiO_2	ILD, passivation, dielectric
SiN_x	PECVD	Si precursor	NH_3 / N_2	SiH_4 + N-containing reactant → SiN_x	Passivation, hard mask, etch stop
SiON	PECVD	Si precursor	N_2O + NH_3/N_2	Si + O + N incorporation	Dielectric / optical film
Epitaxial Si	Epi-CVD	Si precursor	H_2	SiH_4 → crystalline Si	Epitaxial device layers
Doped Poly-Si	LPCVD	Si precursor	PH_3 / B_2H_6	In-situ doped Si	Electrode/resistor structures
HDP SiO_2	HDP-CVD	Si precursor	O_2	Plasma-assisted oxide deposition	Gap fill / dielectric

Why SiH_4 Is Particularly Hazardous

Pyrophoric — spontaneous ignition / fire / explosion → delayed ignition or accumulation can lead to a much more severe explosion

Pyrophoricity	Very High	May spontaneously ignite when exposed to air	Enclosed gas cabinet, automatic shutoff
Fire	Very High	Extremely flammable	Leak detection + ventilation + isolation
Explosion	Very High	Certain releases/mixing conditions can generate explosive mixtures or delayed ignition	Prevent accumulation; engineered vent/abatement
Inhalation	High	Respiratory/eye/skin irritation; exposure must be controlled	Closed system, ventilation, gas monitoring
Asphyxiation	High	Large release can displace oxygen	Ventilation + O_2 monitoring where appropriate
High-pressure cylinder	High	Stored/delivered from pressurized containers	Approved cylinder restraint and gas system
Oxidizer incompatibility	Very High	Violent reaction possible with oxidizing materials	Physical/system segregation

NIOSH identifies silane as a flammable gas that may ignite spontaneously in air and gives a REL of 5 ppm TWA.

2. Tetraethyl Orthosilicate, TEOS, ($Si(OC_2H_5)_4$)

TEOS = liquid silicon precursor → vaporized → CVD reaction with O_2/O_3/plasma → high-quality/conformal SiO_2, with the facility focus shifting from the pyrophoric-gas risk of SiH_4 toward flammable-liquid/vapor control, condensation prevention, and SiO_2 particulate/deposit management.

SiO_2	LPCVD	Si precursor	O_2 / thermal decomposition chemistry	Good conformality and film quality	ILD, dielectric
SiO_2	PECVD	Si precursor	O_2 / plasma	Lower deposition temperature	Passivation, ILD, dielectric
SiO_2	SACVD	Si precursor	O_3	Good conformality and gap fill	STI/gap fill, ILD
O_3-TEOS SiO_2	CVD/SACVD	Si precursor	O_3	High reactivity and good step coverage	Gap-fill dielectric
Doped SiO_2	CVD	Si precursor	O_2/O_3 + B/P precursor	Tunable dielectric properties	BPSG/PSG-type dielectric
SiO_2-like film	Plasma process	Si precursor	O_2 plasma	Lower-temperature deposition	Temperature-sensitive structures

Why use TEOS instead of SiH_4 for SiO_2?

Parameter	TEOS	SiH_4
Material at room temperature	Liquid	Gas
Delivery to chamber	Vaporized liquid	Compressed gas
Pyrophoric	No	Yes – major concern
Flammability	Flammable	Extremely flammable / pyrophoric
SiO_2 conformality	Excellent / very good	Process dependent
Step coverage	Very good	Moderate–good depending on process
Gap filling	Good, especially O_3-TEOS	Process dependent
Carbon contamination concern	Possible due to organic ligands	Low carbon source
Precursor delivery complexity	Bubbler / vaporizer / DLI	Gas delivery
Typical strength	Conformal oxide deposition	High-reactivity Si precursor

Main TEOS Hazards

Flammable liquid	High	TEOS has a relatively low flash point	Approved chemical cabinet / enclosure
Flammable vapor	High	Vapor can form combustible mixtures	Ventilation + ignition-source control
Pyrophoricity	Low	TEOS is not normally classified as pyrophoric	Major advantage vs. SiH_4
Inhalation	Moderate–High	Vapor/mist can cause irritation and harmful exposure	Closed system + local exhaust
Eye contact	High	Can cause significant irritation	Closed handling + appropriate PPE
Skin contact	Moderate	Irritation possible	Prevent direct contact
Moisture reaction	Moderate	Hydrolysis can produce ethanol and Si-containing hydrolysis products	Moisture-controlled delivery
Leak / spill	High	Creates both chemical exposure and flammable-vapor hazard	Spill containment + ventilation
Condensation	Process concern	Can contaminate delivery lines	Temperature-controlled lines
Particulate formation	Process concern	Premature reaction/hydrolysis can generate SiO_2-related particles	

3. Dichlorosilane, DCS, (SiH_2Cl_2)

DCS is another major silicon precursor gas, especially important for LPCVD silicon nitride, epitaxial silicon/SiGe, and some silicon-containing films.

DCS + NH_3 → LPCVD Si_3N_4 $\qquad$ $3SiH_2Cl_2 + 4NH_3 \rightarrow Si_3N_4 + 6HCl + 6H_2$

Film	Deposition	DCS Role	Main Reactant / Gas	Typical Application
Si_3N_4	LPCVD	Si precursor	NH_3	Hard mask, oxidation barrier, spacer, MEMS structural film
Epitaxial Si	Epi-CVD	Si precursor	H_2	Device epitaxy, source/drain structures
SiGe	Epi-CVD	Si precursor	$GeH_4 + H_2$	Strained SiGe, source/drain engineering
Doped Si / SiGe	Epitaxy	Si precursor	PH_3/B_2H_6 etc. + H_2	In-situ doped epi
Poly-Si	LPCVD/CVD	Si precursor	H_2 / thermal decomposition	Gate/electrode/MEMS, process dependent
Si-containing films	CVD	Si precursor	Chemistry-dependent	Specialized dielectric/semiconductor layers

HCl is generated as a process by-product, so the exhaust system must manage both unreacted precursor and corrosive acidic exhaust.

Main Dichlorosilane (DCS) Hazards

Flammability	Very High	DCS is a highly flammable chlorosilane gas	Gas cabinet + automatic isolation
Fire / explosion	Very High	Release can create dangerous combustible mixtures	Leak detection + ventilation + interlocks
Inhalation	High	Harmful gas/exposure products	Closed delivery + gas detection
Moisture reactivity	High	Reacts with moisture	Dry UHP gas system
Corrosivity	High	Moisture/reaction chemistry can produce HCl	Compatible piping + exhaust treatment
High-pressure / liquefied gas	High	Cylinder contains pressurized material	Approved cylinder/gas cabinet
Air exposure	High	Flammability/reactivity	Prevent leakage and accumulation
Exhaust deposits	High	Si-containing solids/by-products can accumulate	Foreline/pump maintenance + abatement

4. Trichlorosilane, TCS, ($SiHCl_3$)

TCS is the most important role is as a silicon precursor for epitaxial silicon growth. It is also a major feedstock for producing high-purity polysilicon.

$$SiHCl_3 + H_2 \rightarrow Si(s) + 3HCl$$

Film / Application	Process	TCS Role	Other Important Gases	Purpose
Epitaxial Si	Epi-CVD	Si precursor	H_2	Grow high-quality single-crystal Si
Doped Epi-Si	Epi-CVD	Si precursor	H_2 + PH_3/AsH_3/B_2H_6	In-situ n/p doping
Thick Epi-Si	Epi-CVD	Si precursor	H_2	Power/discrete device epi
Si surface preparation	Epi reactor	—	H_2 / HCl	Surface cleaning/conditioning before growth
Polysilicon production	Siemens-type process	Si feedstock	H_2	Produce high-purity polysilicon feedstock

Hydrogen acts as both process/carrier gas and reducing environment, while HCl exits as a gaseous reaction product.

TCS Main Hazards

The chlorine chemistry can be useful for epitaxial selectivity/process control, but it also creates the major disadvantage: HCl and chlorinated exhaust make the downstream system highly corrosion-sensitive.

Flammability	Very High	Flammable liquid and vapor	Enclosed delivery + ventilation
Fire / explosion	Very High	Very broad flammable range	Leak detection + automatic isolation
Moisture reactivity	Very High	Reacts with moisture	Dry UHP system
Corrosivity	Very High	Moisture/reaction can generate HCl	Compatible materials + scrubber
Inhalation	High	Vapor/reaction products hazardous	Closed delivery and exhaust
Vapor accumulation	High	Vapor is significantly heavier than air	Ventilated enclosure and leak control
Liquid spill	High	Rapid vapor generation + reaction with moisture	Secondary containment
Exhaust deposits	High	Si-containing material can accumulate	Foreline/pump maintenance

Useful comparison: SiH_4 vs DCS vs TCS vs TEOS

Parameter	SiH_4	DCS	TCS	TEOS
Full Name	Silane	Dichlorosilane	Trichlorosilane	Tetraethyl Orthosilicate
Formula	SiH_4	SiH_2Cl_2	$SiHCl_3$	$Si(OC_2H_5)_4$
Cl atoms	0	2	3	0
Physical form near RT	Gas	Gas / liquefied gas	Volatile liquid	Liquid
Boiling Point	~-112°C	~8°C	~32°C	~168°C
Main Process	LPCVD/PECVD	LPCVD/Epi	Epitaxy	CVD/PECVD/SACVD
Signature Film	Poly-Si / SiO_2 / SiN	Si_3N_4	Epitaxial Si	SiO_2
Main Reactant/Carrier	Process dependent	NH_3/H_2	H_2	O_2/O_3
Major Hazard	Pyrophoric	Flammable/reactive	Flammable + moisture reactive	Flammable liquid
Corrosive Exhaust	Low–moderate	High	Very High	Lower
Major By-product Concern	Si particles/oxide	HCl	HCl	VOC + SiO_2
Delivery Concept	Gas cylinder	Gas/liquefied-gas source	Liquid/vapor delivery	Bubbler/DLI

5. Trimethylaluminum, TMA, ($Al(CH_3)_3$)

It is one of the most important metal-organic precursors in semiconductor manufacturing, particularly for ALD/PEALD deposition of Al_2O_3.

$TMA + H_2O \rightarrow Al_2O_3 + CH_4$

The ALD sequence is:
① **TMA pulse** → ② **N_2/Ar purge** → ③ **H_2O pulse** → ④ **N_2/Ar purge** → **Repeat**

Film	Deposition	TMA Role	Reactant	Main Application
Al_2O_3	Thermal ALD	Al precursor	H_2O	High-k dielectric, passivation
Al_2O_3	Thermal ALD	Al precursor	O_3	High-quality oxide
Al_2O_3	PEALD	Al precursor	O_2 plasma	Low-temperature/high-quality oxide
AlN	ALD/PEALD	Al precursor	NH_3 / N-containing plasma	Dielectric, barrier, piezo/functional layers
Al-containing high-k	ALD	Al precursor	H_2O/O_3 + another metal precursor	HfAlO, Al-doped oxides
Al-containing films	CVD	Al precursor	Process dependent	Specialized applications

Self-limiting reaction → atomic-scale thickness control → excellent conformality
which is extremely valuable for FinFET, GAAFET, high-aspect-ratio structures, capacitors, MEMS and advanced 3D devices.

TMA Main Hazards

Pyrophoric	Extreme	Ignites spontaneously in air	Closed precursor system
Water reactive	Extreme	Violently reacts with water/moisture	Completely dry system
Fire	Extreme	Rapid combustion after release	Automatic isolation + ventilation
Chemical burns	High	Highly reactive material	Prevent exposure
Inhalation	High	Vapor/reaction products hazardous	Closed delivery + exhaust
Liquid leak	Extreme	Can ignite immediately upon air exposure	Secondary containment / engineered enclosure
Air/moisture contamination	Extreme	Can cause reaction inside delivery system	UHP inert purge

6. Trimethylgallium (TMGa), Trimethylaluminum (TMAl), and Trimethylindium (TMIn)

TMGa, TMAl, and TMIn are Group-III metal-organic precursors used in MOCVD/MOVPE, especially for GaN-based compound semiconductors. They supply Ga, Al, and In, while NH_3 typically supplies nitrogen.

GaN	TMGa	NH_3	Buffer/channel, LED, HEMT
AlN	TMAl	NH_3	Nucleation/buffer, barrier
InN	TMIn	NH_3	Specialized III-N structures
AlGaN	TMAl + TMGa	NH_3	GaN HEMT barrier, UV LED
InGaN	TMIn + TMGa	NH_3	Blue/green LED MQW, laser
AlInN	TMAl + TMIn	NH_3	HEMT/barrier structures
AlInGaN	TMAl + TMIn + TMGa	NH_3	Advanced III-N heterostructures
GaAs	TMGa	AsH_3	III-V optoelectronics
AlGaAs	TMAl + TMGa	AsH_3	Laser, HBT, optical structures
InGaAs	TMIn + TMGa	AsH_3	High-mobility devices, photonics

How the MOCVD Chemistry Works

The basic concept is:
Metal-organic precursor vapor + Group-V precursor → heated wafer → epitaxial III-V film

For GaN: TMGa vapor + NH_3 → GaN epitaxial film + hydrocarbon-containing by-products
For AlN: TMAl vapor + NH_3 → AlN epitaxial film + hydrocarbon-containing by-products
For InGaN: TMGa + TMIn + NH_3 → $In_xGa_{1-x}N$

These materials are not conventional compressed process gases like N_2, H_2 or NH_3.
The general delivery concept is:
Metal-organic precursor source
→ temperature-controlled bubbler/source vessel
→ H_2 or N_2 carrier gas
→ precursor vapor
→ MOCVD gas distribution/showerhead
→ heated wafer

TMGa, TMAl, TMIn Main Hazards

	TMGa	TMAl	TMIn
Pyrophoric	● Very High	●● Extreme	● Very High
Flammable	● High	● Very High	● High
Air Sensitive	● Very High	● Extreme	● Very High
Moisture Sensitive	● Very High	● Extreme	● Very High
Fire on Air Exposure	Major concern	Major concern	Major concern
Uncontrolled Water Contact	Dangerous	Especially dangerous	Dangerous
Metal-containing exhaust	Yes	Yes	Yes
Organic/CH_x by-products	Yes	Yes	Yes

7. Tetrakis(ethylmethylamino)hafnium (TEMAH)

A hafnium-containing metal-organic precursor used mainly in ALD and PEALD to deposit HfO_2 (hafnium oxide). HfO_2 is an important high-k dielectric in advanced CMOS.

The most important reaction concept is: TEMAH + oxidant → HfO_2
For thermal ALD: TEMAH pulse → N_2/Ar purge → H_2O or O_3 pulse → N_2/Ar purge → Repeat
For PEALD: TEMAH pulse → purge → O_2 plasma → purge → Repeat

Film	Method	Hf Precursor	Reactant	Typical Application
HfO_2	Thermal ALD	TEMAH	H_2O	High-k gate dielectric
HfO_2	Thermal ALD	TEMAH	O_3	High-quality high-k dielectric
HfO_2	PEALD	TEMAH	O_2 plasma	Low-temperature HfO_2
$HfSiO_x$	ALD	TEMAH + Si precursor	H_2O/O_3	High-k dielectric engineering
$HfAlO_x$	ALD	TEMAH + TMA	H_2O/O_3	High-k / dielectric engineering
HfN	ALD/PEALD	TEMAH	NH_3 / N-H plasma	Specialized electrode/barrier applications

For advanced CMOS, TEMAH → HfO_2 → High-k Metal Gate (HKMG) → FinFET/GAAFET is the key connection.

TEMAH Handling & Hazards

Hazard / Issue	Concern	Typical Control
Air sensitivity	Precursor degradation/reaction	Closed inert delivery
Moisture sensitivity	Reacts/degrades with moisture	Dry UHP system
Combustibility	Organic precursor presents fire hazard	Ventilated precursor enclosure
Vapor exposure	Harmful exposure should be avoided	Closed delivery + exhaust
Thermal decomposition	Excessive heating can degrade precursor	Precise source/line temperature control
Condensation	Causes unstable delivery	Heated delivery lines
Contamination	C/N residues can affect film quality	Optimized ALD chemistry/purge
Hf-containing deposits	Exhaust/foreline contamination	Abatement + PM

8. Tetrakis(dimethylamido)hafnium(IV), TDMAH, ($Hf[N(CH_3)_2]_4$)

It is a hafnium-containing metal-organic/metal-amide precursor used mainly in ALD and PEALD to deposit HfO_2 high-k dielectric films.

Film	Hf Precursor	Reactant	Method	Application
HfO_2	TDMAH	H_2O	ALD	High-k gate dielectric
HfO_2	TDMAH	O_3	ALD	High-quality HfO_2
HfO_2	TDMAH	O_2 plasma	PEALD	Low-temperature/high-reactivity deposition
$HfSiO_x$	TDMAH + Si precursor	Oxidant	ALD	Engineered high-k
HfN	TDMAH	NH_3/N-H plasma	ALD/PEALD	Specialized electrode/barrier

Handling and Hazards of TDMAH

Issue	Concern	Typical Control
Air exposure	Precursor reaction/degradation	Closed delivery
Moisture	Unwanted reaction/decomposition	Dry UHP system
Heating	Needed for stable vapor delivery	Temperature-controlled vessel
Overheating	Thermal decomposition	Precise temperature control
Condensation	Unstable precursor dose / line blockage	Heated lines
Vapor exposure	Chemical exposure hazard	Enclosed precursor cabinet
C/N contamination	Can affect HfO_2 electrical properties	Optimize ALD conditions
Foreline deposits	Maintenance/particle issue	Heated foreline + PM
Exhaust precursor	Reactive chemical	Compatible abatement

Comparison of TMA vs TDMAH vs TEMAH

TDMAH/TEMAH → HfO_2 → HKMG → FinFET → GAAFET

TMA → Al → Al_2O_3 → dielectric

Metal	Al	Hf	Hf
Main film	Al_2O_3	HfO_2	HfO_2
Precursor family	Alkyl	Metal-amide	Metal-amide
Main process	ALD/PEALD	ALD/PEALD	ALD/PEALD
Oxidant	$H_2O/O_3/O_2$ plasma	$H_2O/O_3/O_2$ plasma	$H_2O/O_3/O_2$ plasma
Main application	Dielectric/passivation	⭐ High-k gate	⭐ High-k gate
Typical delivery challenge	Very reactive precursor	Heating/vapor control	Heating/vapor control
Important impurities	C	C/N	C/N

9. Ammonia (NH_3)

NH_3 (ammonia) is an important **nitrogen-containing specialty gas** in semiconductor manufacturing. Its primary role is as a **nitrogen source for deposition and nitridation**, especially for forming **SiN_x, SiON, TiN, and other nitride films**.

Process	NH_3 Function	Other Precursor / Gas	Result / Film	Typical Application
LPCVD Si_3N_4	Nitrogen source	DCS (SiH_2Cl_2)	Si_3N_4	Spacer, mask, dielectric, MEMS
LPCVD Si_3N_4	Nitrogen source	SiH_4	SiN_x	Dielectric/passivation
PECVD SiN_x	Nitrogen source	SiH_4 + N_2/NH_3	SiN_x:H	Passivation, dielectric, etch stop
PECVD SiON	Nitrogen source	SiH_4 + N_2O/O_2	SiON	Optical/dielectric layers
ALD TiN	N reactant	$TiCl_4$ or metal-organic Ti precursor	TiN	Barrier, liner, electrode
CVD TiN	Nitrogen source	$TiCl_4$ / Ti precursor	TiN	Metal barrier
ALD/CVD metal nitrides	N reactant	Metal precursor	TaN, WN, etc.*	Barrier/liner/electrode
Thermal nitridation	Nitrogen incorporation	—	Nitrided surface	Dielectric/interface engineering
GaN MOCVD	Nitrogen precursor	TMGa	GaN	LED, RF, power devices
AlGaN MOCVD	Nitrogen precursor	TMGa + TMAl	AlGaN	HEMT / RF / power
AlN MOCVD	Nitrogen precursor	TMAl	AlN	Buffer, piezoelectric/compound semiconductor
InGaN MOCVD	Nitrogen precursor	TMGa + TMIn	InGaN	LED / optoelectronics

NH_3 vs. other nitrogen-source gases

Why NH_3 instead of N_2? → reason is **reactivity**.
N_2: N≡N has a very strong triple bond, → difficult to dissociate → relatively inert.
NH_3: N–H bonds can be thermally or plasma activated more readily, → produces reactive nitrogen-containing species → facilitates nitride-film formation.
N_2 = mainly inert/protective gas; NH_3 = reactive nitrogen precursor.

N_2	Stable/inert N source	Purge, reactive PVD, some plasma processes	Very stable N≡N bond
NH_3	Highly useful reactive N source	SiN, TiN, GaN, AlN	Excellent nitridation/deposition reactant
N_2O	N + O source	SiON / dielectric processing	Oxidizing + nitriding
NO	N + O source	Interface annealing	Interface engineering, especially SiC
N_2 plasma	Activated N species	Plasma ALD / nitridation	Higher reactivity than molecular N_2
NH_3 plasma	Activated N/H species	PEALD / PECVD	High reactivity at lower substrate temperature

Gas Used in Metallization Process

Gases Used in Semiconductor Metallization

Gas / Precursor	Full Name	Key Role	Typical Process	Target Film / Application
Ar	Argon	Sputtering plasma / physical bombardment	PVD sputtering	Al, Cu, Ti, Ta, W, TiN, TaN
N_2	Nitrogen	Reactive gas, purge, inert atmosphere	Reactive PVD, anneal, CVD purge	TiN, TaN, nitrides; annealing
H_2	Hydrogen	Reducing gas / surface cleaning	CVD, anneal, plasma treatment	W-CVD, oxide reduction, surface preparation
NH_3	Ammonia	Nitrogen source / reducing reactant	CVD / ALD	TiN, TaN, WN and other nitrides
WF_6	Tungsten Hexafluoride	Tungsten precursor	CVD-W / ALD-W	W plug, contact/via fill
SiH_4	Silane	Reducing agent	CVD-W nucleation	W deposition from WF_6
B_2H_6	Diborane	Strong reducing/nucleation agent	CVD/ALD-W	W nucleation / low-resistivity W processes
H_2 + WF_6	Hydrogen + Tungsten Hexafluoride	Reduction chemistry	CVD	Bulk W deposition
$TiCl_4$	Titanium Tetrachloride vapor	Ti-containing precursor	CVD / ALD	Ti / TiN barrier/liner
TDMAT	Tetrakis(dimethylamido)titanium	Metal-organic Ti precursor	MOCVD / ALD	TiN barrier
TDEAT	Tetrakis(diethylamido)titanium	Metal-organic Ti precursor	MOCVD / ALD	TiN
Ta-containing precursors	e.g., PDMAT-type chemistry	Ta source	ALD / CVD	Ta/TaN Cu diffusion barrier
Co precursors	Metal-organic Co compounds	Co source	CVD / ALD	Co liner/contact/interconnect
Ru precursors	Metal-organic Ru compounds	Ru source	CVD / ALD	Ru liner / advanced interconnect
NF_3	Nitrogen Trifluoride	Chamber cleaning	Remote-plasma clean	Removes deposition residues
F_2	Fluorine	Chamber cleaning	CVD chamber clean	W/metal deposition chamber cleaning

Gas usage by metallization process

1. **PVD / Sputtering: Ar + N_2** → Al, Cu, Ti/TiN, Ta/TaN
2. **Tungsten CVD: WF_6 + H_2 / SiH_4 / B_2H_6** → W contact & via
3. **Barrier / Liner CVD-ALD: $TiCl_4$, NH_3, metal-organic Ti/Ta/Co/Ru precursors** → TiN, TaN, Co, Ru
4. **Pre-clean & Annealing: Ar, H_2, N_2, Ar/H_2, forming gas** → oxide removal, reduction, thermal treatment
5. **Metal Etching: Cl_2 + BCl_3 + Ar** → particularly Al/AlCu and Ti/TiN
6. **Chamber Cleaning: NF_3 / F_2** → remove deposition residues.

Metallization Step	Main Gases / Vapors	Function
Pre-clean	Ar, H_2, Ar/H_2	Remove native oxide / contamination
PVD Ti/TiN	Ar, N_2	Sputtering + reactive nitridation
PVD Ta/TaN	Ar, N_2	Cu barrier deposition
PVD Al / AlCu	Ar	Metal sputtering
PVD Cu seed	Ar	Cu seed deposition before electroplating
CVD/ALD TiN	$TiCl_4$ + NH_3 or metal-organic precursor	Barrier/liner deposition
CVD-W	WF_6 + H_2 / SiH_4 / B_2H_6	W contact/via fill
Cu electroplating	Normally no process gas	Cu deposited from liquid electrolyte
Annealing	N_2, Ar, H_2/N_2	Grain improvement, reduction, stabilization
Al metal etch	Cl_2 + BCl_3, sometimes Ar	Pattern Al/AlCu
Chamber cleaning	NF_3, F_2	Remove chamber deposits
Advanced Co/Ru deposition	Metal-organic precursor + H_2/NH_3/etc.	

Gas Used in Thermal Process

Functional Groups of Gases in Thermal Process

Thermal processes use several gases depending on whether the objective is oxidation, dopant activation, metal sintering, interface passivation, nitridation, or simply providing an inert high-temperature ambient.

Gas Group	Main Gases	Function
① Inert / Protective	N_2, Ar	Prevent unwanted oxidation; purge; high-T ambient
② Reducing / Passivation	H_2, N_2/H_2, Ar/H_2	Oxide reduction, interface passivation
③ Oxidation	O_2, H_2O vapor	Dry/wet thermal SiO_2 growth
④ Nitridation / Oxynitridation	NH_3, N_2O, NO	Nitrogen incorporation and interface engineering
⑤ Specialized Reactive	HCl and related chemistries	Surface/contamination control in specialized processes

Gas selection by thermal process

Thermal Process	Typical Gas	Main Purpose	Typical Temperature*	Example Application
RTA – Dopant Activation	N_2 / Ar	Activate implanted dopants; repair implant damage	~800–1,100°C	CMOS source/drain
RTP – Oxidation (RTO)	O_2 / O_2-N_2	Rapid thin oxide growth	~800–1,100°C	Thin SiO_2/interface oxidation
RTA – Silicidation	N_2 / Ar	Promote metal–Si reaction without unwanted oxidation	~300–800°C	NiSi, $CoSi_2$, $TiSi_2$
Metal Annealing	N_2 / Ar / forming gas	Reduce stress, improve film/contact properties	~200–500°C+	Al, Cu, contacts
Forming Gas Anneal — FGA	N_2/H_2	Passivate Si/SiO_2 interface defects	~350–450°C	CMOS / MOS devices
Sintering	N_2/H_2 or N_2	Improve metal/Si contact and interface quality	~350–500°C	Al-based metallization/contact
Dopant Drive-in	N_2 / O_2	Redistribute dopants by diffusion	~900–1,200°C	Traditional junction formation
Dry Oxidation	O_2	High-quality SiO_2	~900–1,200°C	Gate/field oxide
Wet Oxidation	H_2O vapor	Fast oxide growth	~900–1,100°C	Thick field/isolation oxide
Oxynitridation	N_2O / NO	Introduce N into oxide/interface	~800–1,100°C	Gate dielectric
SiC Activation Anneal	Ar	Activate implanted dopants and repair lattice damage	~1,500–1,800°C	SiC MOSFET
SiC Interface Anneal	NO / N_2O	Improve SiO_2/SiC interface quality	~1,000–1,300°C	SiC MOSFET gate oxide
High-T Inert Anneal	Ar / N_2	Recrystallization / defect repair	Process dependent	Si, SiC, compound semiconductor

*Recipe- and technology-dependent rather than a universal thermal-process gas.

10. Forming Gas (FG)

Forming gas is a controlled mixture of hydrogen (H_2) diluted in an inert gas, most commonly nitrogen (N_2). In semiconductor manufacturing, it is primarily used to create a mild reducing and passivating atmosphere during thermal treatment.

H_2 + N_2, commonly around 5–10% H_2 with balance N_2; exact concentration is fab/process dependent (e.g. H_2/4.9%N_2 FG or H_2/4.9%Ar FG)

Why mix H_2 with N_2?

Pure H_2 is highly flammable and often provides more reducing capability than needed. Diluting it with N_2 gives a more controllable process atmosphere while retaining useful hydrogen chemistry.

The roles are essentially:

N_2 → dilution + inert carrier/protective atmosphere

H_2 → reduction + hydrogen passivation

The most important semiconductor mechanism is **hydrogen passivation**.

At a Si/SiO_2 interface, structural imperfections can leave unsatisfied Si bonds, often represented conceptually as: **Si• + H → Si–H**

Hydrogen terminates these **dangling bonds**, reducing electrically active interface traps.

Major Applications of Forming Gas

Application	Typical Purpose	Main Effect
MOS Forming Gas Anneal (FGA)	Passivate Si/SiO_2 interface defects	↓ Interface trap density (D_{it})
Post-gate dielectric deposition anneal (PDA)	Improve dielectric/interface quality	Better MOS electrical characteristics
Post-metallization anneal (PMA)	Passivate interface defects after BEOL thermal processing	Improve device performance/reliability
Al sintering	Improve Al/contact interface and contact properties	↓ Contact resistance
Contact annealing	Improve metal-semiconductor interface	Lower resistance / better contact stability
Surface oxide reduction	Reduce certain metal surface oxides	Cleaner metal surface
Sensor/MEMS processing	Passivate Si-based interfaces/surfaces	Improve electrical stability
Solar-cell processing	Hydrogen passivation of electrically active defects	Reduce recombination
Some packaging processes	Provide reducing atmosphere during heating	Reduce m

Forming Gas vs. Other Annealing Gases

Forming Gas = H_2 diluted in N_2, used as a reducing and hydrogen-passivating atmosphere during semiconductor thermal processing for interface-defect passivation and post-metallization annealing
N_2 = Protect → Ar = Inert → O_2 = Oxidize → H_2 = Reduce → Forming Gas = Reduce + Passivate.

N_2/H_2 = Passivation-focused forming gas **Ar/H_2 = Highly inert + reducing forming gas**

Gas / Mixture	Atmosphere	Primary Purpose	Typical Applications	Key Advantage
N_2	Inert / protective	Prevent unwanted oxidation; purge	RTA/RTP, dopant activation, metal anneal, silicidation	Low cost, widely available
Ar	Highly inert	Provide chemically inert high-temperature ambient	RTA, SiC activation, metal anneal, high-T processes	Very low chemical reactivity
H_2	Strong reducing	Oxide reduction, surface cleaning, hydrogenation	Specialized anneal, surface treatment	Strong reducing capability
N_2/H_2 Forming Gas	Mild reducing + passivating	Hydrogen passivation; oxide reduction	FGA, MOS interface passivation, post-metal anneal, sintering	Economical; excellent for Si/SiO_2 defect passivation
Ar/H_2 Forming Gas	Inert + reducing/passivating	Surface oxide reduction, hydrogen passivation, surface cleaning	Metal/contact anneal, compound semiconductors, specialized RTA/RTP, surface treatment	More chemically inert carrier; avoids N_2-related reactions
O_2	Oxidizing	Oxide growth / oxygen annealing	Thermal oxidation, RTO, oxide densification	Controlled oxidation
O_2/N_2	Controlled oxidizing	Control oxygen partial pressure	RTO, interface engineering	Better oxidation-rate control
NO / N_2O	Oxidizing + nitriding	Interface passivation / nitrogen incorporation	Si/SiO_2 and especially SiC/SiO_2 interface engineering	Improves selected dielectric interfaces
NH_3	Nitriding / reducing-reactive	Nitridation	Surface nitridation, nitride-related thermal processes	Strong nitrogen

Gas Used in Diffusion & Ion Implantation Process

11. Arsine (AsH_3)

An arsenic hydride specialty gas used in semiconductor manufacturing mainly as an arsenic source for n-type doping and as a Group-V precursor for III-V epitaxial growth.

1. Silicon doping: AsH_3 provides arsenic → n-type Si
AsH_3 → As source → ionization → As^+ implantation → annealing → electrically active n-type Si

2. III-V epitaxy: TMGa + AsH_3 → GaAs

Ion Implantation	As dopant source	As^+ ions implanted into Si	n-type source/drain, well/contact regions
Si Epitaxy	In-situ dopant	As-doped epitaxial Si	n-type epi
SiGe Epitaxy	In-situ dopant	As-doped SiGe	Advanced CMOS structures
GaAs MOCVD	Group-V As precursor	GaAs	RF, photonics, optoelectronics
AlGaAs MOCVD	As precursor	AlGaAs	Laser, HBT, heterostructure
InGaAs MOCVD	As precursor	InGaAs	High-speed electronics, photodetector
InAlGaAs etc.	As precursor	III-V alloys	Advanced heterostructures

Why AsH_3 is particularly dangerous

Acute toxicity	Extreme	Very small exposures can cause severe systemic poisoning	Closed system + continuous detection
Blood toxicity	Extreme	Can cause massive hemolysis (destruction of red blood cells)	Prevent exposure entirely
Kidney injury	High	Can result from severe hemolysis	Exposure prevention
Inhalation	Critical	Primary exposure route	Ventilated gas cabinet
Flammability	High	Flammable gas	Ignition control/isolation
Gas leakage	Critical	Invisible toxic release	Toxic gas detector + automatic shutoff
Arsenic contamination	High	Toxic residues/by-products	Controlled maintenance/waste handling

12. Phosphine (PH_3)

It is primarily a phosphorus source for n-type doping of silicon and a Group-V precursor for certain III-V semiconductor materials.

For ion implantation: PH_3 → ion source → ionization → mass selection → acceleration → wafer → annealing
For In-Situ Doping: SiH_4 / DCS + PH_3 → phosphorus-doped Si
PH_3 in III-V Epitaxy: InPTMIn + PH_3 → InP

Ion Implantation	P source	P^+/P-containing ions	n-type doping of Si
Diffusion	P source	Phosphorus-doped Si	Junction / well formation, legacy processes
Si Epitaxy	In-situ n-type dopant	P-doped epi-Si	Epitaxial device layers
SiGe Epitaxy	n-type dopant	P-doped SiGe	Advanced CMOS
Poly-Si LPCVD	In-situ dopant	n^+ poly-Si	Gate/electrode/resistor
CVD	Dopant source	P-containing/doped films	Process-dependent
III-V MOCVD	Group-V P precursor	InP, GaP etc.	Photonics / high-speed devices

Main PH_3 Hazards

Acute toxicity	Extreme	Very toxic by inhalation	Closed gas system
Inhalation	Critical	Can cause severe systemic/respiratory effects	Continuous GDS
Flammability	High	Flammable gas	Ventilation + isolation
Fire / ignition	High	Certain concentrations/conditions can ignite	Automatic shutoff
Gas leakage	Critical	Toxic release into fab	Gas cabinet + GDS
High-pressure cylinder	High	Stored as pressurized material	Approved cylinder system
Process exhaust	High	Residual PH_3 remains toxic	Dedicated abatement
Maintenance residue	High	P-containing deposits may remain hazardous/reactive	Controlled PM procedures

13. Diborane (B_2H_6)

Diborane (B_2H_6) is a key boron-source electronic specialty gas used primarily for p-type doping of silicon, SiGe, polysilicon, and epitaxial layers. Compared with PH_3 and AsH_3, the most important distinction is: B_2H_6 → B → p-type, whereas PH_3 / AsH_3 → P / As → n-type.

Ion Implantation	B source	Boron-containing ion beam	p-type Si doping
Si Epitaxy	In-situ p-type dopant	B-doped epi-Si	Source/drain, device layers
SiGe Epitaxy	p-type dopant	B-doped SiGe	pMOS source/drain, advanced CMOS
Poly-Si LPCVD	In-situ dopant	p^+ poly-Si	Gate/electrode/resistor structures
CVD	Boron source	B-doped films	Specialized processes
B-containing dielectric deposition	Boron precursor	B-containing glass/film	Process-dependent
III-V epitaxy	Dopant	B incorporation / specialized layers	Less common than its Si doping role

B_2H_6 Main Hazards

B_2H_6 combines high toxicity with severe flammability/reactivity, it is treated as a high-hazard specialty gas requiring multiple independent engineering controls.

Acute toxicity	Extreme	Highly toxic by inhalation	Closed delivery system
Flammability	Extreme	Highly flammable gas	Ventilated enclosure
Fire / ignition	Very High	Can ignite readily	Automatic isolation
Gas leakage	Critical	Toxic + fire hazard simultaneously	GDS + emergency shutoff
Inhalation	Critical	Severe respiratory/systemic effects possible	Continuous detection
High-pressure source	High	Cylinder hazard	Approved gas system
Exhaust	High	Residual B_2H_6 remains hazardous	Dedicated abatement
Process residues	–	Boron-containing material	Controlled maintenance

14. Boron Trifluoride (BF_3)

A boron-source gas for p-type ion implantation. Unlike diborane (B_2H_6), BF_3 is nonflammable, but it is toxic, corrosive, and highly reactive with moisture.

BF_3 → B source → ionization → B-containing ion beam → Si wafer → p-type Si

Application	BF_3 Role	Purpose
Ion implantation	⭐ Boron source	p-type doping
p-well formation	B dopant	CMOS well formation
p-type source/drain	B dopant	PMOS structures
Threshold-voltage adjustment	Low-dose B implantation	Vt control
Channel engineering	B source	Device profile engineering
Junction formation	B implantation	p-n junction
Specialized implants	B/F-containing ion species	Process-dependent

Handling & Safety of BF_3

Hazard	Severity	Engineering Control
Toxic inhalation	High	Closed delivery + ventilation
Corrosive gas	High	Compatible gas system
Moisture reaction	High	Dry UHP system
HF formation in wet conditions	High	Moisture exclusion + appropriate abatement
Cylinder leak	High	Gas cabinet + automatic isolation
Process exhaust	High	Dedicated treatment
Flammability	Not flammable	Advantage vs B_2H_6
Pyrophoricity	Not pyrophoric	Advantage vs B_2H_6

Comparison of BF_3 vs B_2H_6 vs PH_3 vs AsH_3

Parameter	BF_3	B_2H_6	PH_3	AsH_3
Full name	Boron Trifluoride	Diborane	Phosphine	Arsine
Formula	BF_3	B_2H_6	PH_3	AsH_3
CAS No.	7/2/7637	19287-45-7	7803-51-2	7784-42-1
Element supplied	B	B	P	As
Periodic group of dopant	Group 13 (III)	Group 13 (III)	Group 15 (V)	Group 15 (V)
Si doping type	🔴 p-type	🔴 p-type	🔵 n-type	🔵 n-type
Electrical behavior	Acceptor	Acceptor	Donor	Donor
Physical state @ RT	Gas	Gas	Gas	Gas
Molecular weight	67.8	27.7	34	78.0 g/mol
Boiling point	~-100°C	~-92.5°C	~-87.7°C	~-62.5°C
Relative gas density, air=1	2.38	0.97	1.18	2.69
Flammable	❌ No	🔴 Yes	🔴 Yes	🔴 Yes
Pyrophoric / spontaneous ignition concern	❌ No	🔴 Very high	🔴 Possible spontaneous ignition	Flammable; ignition control required
Toxicity concern	🔴 High	🔴🔴 Very high	🔴🔴🔴 Extreme	🔴🔴🔴 Extreme
Corrosive/moisture chemistry	🔴 High	Reactive	Less dominant	Less dominant
Main unique hazard	BF_3 + moisture → HF-containing chemistry	Toxic + pyrophoric/flammable	Extremely toxic + flammable	Extreme systemic/hemolytic toxicity
NIOSH IDLH	25 ppm	15 ppm	50 ppm	3 ppm
Main semiconductor use	⭐ Ion implantation	CVD/epi doping + implant	Implant + CVD/epi doping	Implant + III-V epitaxy
Typical purge gas	UHP N_2	UHP N_2	UHP N_2	UHP N_2
Gas cabinet / controlled enclosure	Yes	Yes	Yes	Yes
Gas detection	Required per facility/code design	Critical	Critical	Critical
Dedicated abatement	Yes	Yes	Yes	Yes

Comparison of BF_3 vs B_2H_6 vs PH_3 vs AsH_3

BF_3 & B_2H_6 → BORON → p-type | PH_3 & AsH_3 → PHOSPHORUS / ARSENIC → n-type

BF_3 → mainly Implant | B_2H_6 → Implant/Epi/CVD | PH_3 → Implant/Epi + InP | AsH_3 → Implant/Epi + GaAs

	BF_3	B_2H_6	PH_3	AsH_3
Element	B	B	P	As
Doping	🔴 p-type	🔴 p-type	🔵 n-type	🔵 n-type
Main role	Implant	Epi/CVD doping	Implant + Epi	Implant + III-V
Flammable	❌	🔥 Yes	🔥 Yes	🔥 Yes
Key hazard	Toxic + corrosive	Toxic + pyrophoric	Extremely toxic + flammable	Extremely toxic
Special issue	HF/fluoride chemistry	Spontaneous ignition	Toxic leak/fire	Hemolysis / As contamination
Other major role	—	B-doped Si/SiGe	InP precursor	GaAs precursor
Memory rule	B → p	B → p	P → n	As → n

Gas Used in Lithography Process

Key Lithography Gas Groups

Wafer → N_2/HMDS Prime → Coat → Bake → ArF/KrF Exposure → PEB → Develop → O_2 Descum → Pattern Ready for Etch/Implant

Group	Main Gases	Key Purpose
1. Purge / Utility	N_2, CDA, He	Clean environment, purge, pneumatics
2. Surface Preparation	HMDS vapor + N_2	Improve photoresist adhesion
3. DUV Exposure	Ar + F_2 + Ne	ArF laser → 193 nm
	Kr + F_2 + Ne	KrF laser → 248 nm
4. EUV Support	H_2	Source/collector contamination management
5. Descum / Resist Removal	O_2, Ar/O_2, CF_4/O_2	Remove residual photoresist and organic contamination

Gases Used in Semiconductor Lithography

Gas	Full Name / Composition	Lithography Area	Main Function	Why It Is Used
N_2	Nitrogen	Track, scanner, coat/develop, bake	Purge, inerting, wafer drying, chemical dispense blanket	Inert, clean, low moisture/O_2; prevents contamination and oxidation
CDA	Clean Dry Air	Track / automation	Pneumatic operation, environmental control	Clean, dry source for valves, actuators and equipment
Ar	Argon	Excimer laser / plasma processes	Laser gas component; inert plasma	Chemically inert; supports laser discharge and plasma
Ne	Neon	ArF/KrF excimer laser	Buffer gas in laser mixture	Efficient energy transfer and stable laser discharge
He	Helium	Scanner / optical system	Purging, thermal management, specialized optical environments	High thermal conductivity and chemically inert
H_2	Hydrogen	EUV scanner/source	EUV source/collector cleaning and contamination control	Helps remove carbon contamination and mitigate Sn-related contamination
Ar + F_2 + Ne	Argon + Fluorine + Neon	ArF excimer laser	Generates 193 nm exposure light	Standard DUV wavelength for advanced lithography
Kr + F_2 + Ne	Krypton + Fluorine + Neon	KrF excimer laser	Generates 248 nm exposure light	Widely used for older/mature technology nodes
F_2	Fluorine	Excimer laser mixture	Active halogen component	Reacts with Ar/Kr excited species to form excimer emission
O_2	Oxygen	Resist strip / descum	O_2 plasma removes residual photoresist	Converts organic resist into volatile products
O_2 + CF_4	Oxygen + Carbon Tetrafluoride	Descum / residue removal	Enhanced plasma cleaning	Fluorine assists removal of difficult residues
CO_2	Carbon Dioxide	Specialized wafer cleaning / lithography support	Cleaning, surface treatment; sometimes dissolved in UPW	Can control electrostatic charging and assist cleaning
NH_3	Ammonia	Specialized resist/process chemistry	Chemical treatment / chemically amplified resist-related processes	Used only in particular processes, not universal
HMDS vapor + N_2	Hexamethyldisilazane carried by N_2	Vapor prime	Improves photoresist adhesion to SiO_2/oxide surfaces	Converts hydrophilic surface to more hydrophobic surface

Gas Used in Plasma Etching Process

Fundamental Principles of Plasma Etching

A material can be etched efficiently only if the surface reaction produces a by-product that can leave the wafer surface under the process conditions.

Etchant species + wafer material → volatile reaction product → desorption → vacuum pump

If the product is nonvolatile, it stays on the surface, causing residue, micromasking, roughness, etch stop, or requiring stronger ion bombardment / higher temperature.

1. Silicon etched with fluorine: $Si(s) + 4F\cdot \rightarrow SiF_4(g)$
SiF_4 is highly volatile, so it readily leaves the wafer.
Therefore: F radical → reacts with Si → SiF_4 gas → pumped away
This is why fluorine chemistry can produce a very high Si etch rate.

2. If Material + Etchant → nonvolatile solid
the product remains on the wafer:
Reaction → surface residue → blocks fresh etchant → etch slows/stops

Thus, selecting F, Cl, or Br is partly about asking:
Which halogen forms the most removable product with the material I want to etch?

Volatility Determines Etchability

Low volatility is not always bad. Controlled sidewall deposition/passivation is deliberately used to obtain vertical profiles.

Situation	Surface Result	Etching Consequence
Highly volatile product	Product readily desorbs	⭐ Fast chemical etch
Moderately volatile	Needs heating/ion assistance	Controlled etch
Low volatility	Product remains longer	Slower etch
Nonvolatile product	Surface residue	❌ Etch inhibition
Nonvolatile + ion bombardment	Product physically removed	Ion-assisted etching possible
Nonvolatile residue on sidewall	Sidewall protected	⭐ Can be useful for anisotropy

The Complete Selection Logic for Etching Gas

If the etch product is volatile → it leaves → etching continues.
If the etch product is nonvolatile → it stays → etching slows/stops, unless ions or heat remove it.

This concept explains why SF_6 is excellent for fast Si etching, HBr is excellent for precise Si etching, Cl_2/BCl_3 is favored for many metals and III-Vs, and C_4F_8 deliberately creates nonvolatile polymer for sidewall protection.

1. Will F, Cl or Br react with the material?	Reaction must be chemically possible
2. What product will form?	SiF_4? $AlCl_3$? WF_6? $InCl_x$?
3. Is the product volatile?	Determines whether it can leave the wafer
4. At what temperature is it removable?	Determines wafer/chamber temperature
5. Is ion bombardment required?	Determines bias/RIE/ICP requirement
6. What is the mask reaction?	Determines selectivity
7. What happens on the sidewall?	Determines anisotropy
8. Are nonvolatile residues formed?	Determines particles/roughness
9. What are the exhaust products?	Determines abatement
10. Does the chemistry meet CD/profile requirements?	Determines final recipe

Etching Gas by Wafer & Film Materials

Plasma etching uses reactive gases that are ionized/dissociated in a plasma to produce radicals and ions that chemically react with—and often physically bombard—the wafer surface

Si	SF_6, CF_4, Cl_2, HBr	F / Cl / Br	Rate vs anisotropy
Poly-Si	⭐ HBr + Cl_2 + O_2	Br/Cl	CD/profile control
SiO_2	CF_4, CHF_3, C_4F_8, CH_2F_2, C_4F_6	Fluorocarbon	Polymer-assisted anisotropy
Si_3N_4	CF_4, CHF_3, CH_2F_2, CH_3F	Fluorocarbon	Selectivity to oxide/Si
Low-k dielectric	CF_4, CHF_3, C_4F_8, C_4F_6 + O_2/Ar	Fluorocarbon	Damage/CD/residue
Photoresist	⭐ O_2	Oxygen plasma	Ashing/strip
Amorphous carbon	O_2, CO_2, H_2/N_2	Oxidation/reduction	Hard-mask removal
Al	⭐ Cl_2 + BCl_3	Chlorine	Volatile $AlCl_x$ products
Ti / TiN	Cl_2, BCl_3, CF_4/SF_6	Cl/F	Metal etch
W	SF_6, NF_3, CF_4	Fluorine	WF_x formation
Ta / TaN	Cl_2, BCl_3, CF_4	Cl/F	Barrier etch
GaN	⭐ Cl_2 + BCl_3 + Ar	Chlorine ICP	III-N dry etching
AlGaN	$Cl_2/BCl_3/Ar$	Chlorine	HEMT structures
GaAs	Cl_2, BCl_3, HBr	Halogen	III-V etching
InP	Cl_2, CH_4/H_2, HBr	Cl/hydrocarbon	Volatility/profile issues
SiC	⭐ SF_6 / NF_3 + O_2/Ar	Fluorine	Strong Si-C bonds
Organic/polymer	⭐ O_2, O_2/CF_4	Oxygen	Ashing/descum

15. Fluorine-based gases — F chemistry

CF_4	Carbon Tetrafluoride	Tetrafluoromethane	SiO_2, SiN_x, Si etching
CHF_3	Trifluoromethane	Fluoroform	SiO_2/SiN_x selective etching, contact/via etch
CH_2F_2	Difluoromethane	Methylene Fluoride	SiN_x/SiO_2 etching; polymer/passivation control
CH_3F	Fluoromethane	Methyl Fluoride	Selective dielectric etching; strong polymer-forming chemistry
C_2F_6	Hexafluoroethane	—	SiO_2/SiN_x etching; plasma cleaning
C_3F_8	Octafluoropropane	Perfluoropropane	Dielectric etching and sidewall passivation
C_4F_8	Octafluorocyclobutane	Perfluorocyclobutane, c-C_4F_8	Bosch DRIE passivation; HAR dielectric etching
C_4F_6	Hexafluoro-1,3-butadiene	Hexafluorobutadiene	Advanced HAR SiO_2/low-k dielectric etching
SF_6	Sulfur Hexafluoride	—	High-rate Si etching, MEMS/TSV DRIE
NF_3	Nitrogen Trifluoride	—	Mainly chamber cleaning; F-radical source
F_2	Fluorine	Molecular Fluorine	Specialized etching and chamber cleaning

Role of C/F ratio in fluorocarbon plasma etching

C/F ratio is one of the most useful ways to understand the balance between chemical etching and fluorocarbon-polymer deposition/passivation.

More F → stronger chemical etching.
More C relative to F → more CF_x polymer → stronger passivation and anisotropy

C/F condition	Plasma behavior	Polymer formation	Etch tendency	Typical result / use
Very low C/F	F-rich	Very low	Strong chemical etching	High etch rate, less sidewall protection
Low C/F	F-rich	Low	Etching dominates	Si etching, oxide etching with lower polymer
Medium C/F	Balanced F + CF_x	Medium	Etch + passivation	Anisotropic dielectric etching
High C/F	C-rich	High	Polymer/passivation increases	High-selectivity oxide etching, sidewall protection
Very high C/F	Strongly polymerizing	Very high	Deposition may compete with etching	Etch stop, chamber polymer buildup possible

Actual plasma chemistry also depends strongly on RF power/bias, pressure, source power, gas dilution, O_2 addition, residence time, chamber wall condition, and ion energy.

Fluorine-based gases — F chemistry

(CF_4, CHF_3, C_2F_6, C_3F_8, C_4F_8, C_4F_6, SF_6, NF_3, F_2)

Gas	Formula	Nominal C/F ratio	Plasma characteristic	Polymer tendency	Main etch target	Typical applications
CF_4	CF_4	0.25	F-rich fluorocarbon	★ Low	SiO_2, SiN_x, Si	General dielectric etch; often mixed with O_2, H_2, Ar
CHF_3	CHF_3	0.33	More CF_x / less free F than CF_4	★★★ High	SiO_2, SiN_x	Oxide/contact etch; improved Si/SiO_2 selectivity and sidewall passivation
C_2F_6	C_2F_6	0.33	F-rich CF_x chemistry	★★ Low–medium	SiO_2, SiN_x	Dielectric etch and historically chamber cleaning
C_3F_8	C_3F_8	0.375	More polymerizing	★★★ Medium–high	SiO_2	Anisotropic dielectric etch
C_4F_8	C_4F_8	0.5	Strong CF_x polymer precursor	★★★★ Very high	SiO_2; Si in cyclic processes	High-AR (HAR) dielectric etch; Bosch DRIE passivation
C_4F_6	C_4F_6	0.67	Highly carbon-rich	★★★★★ Very high	SiO_2 / low-k	Advanced HAR oxide etch; high selectivity/profile control
SF_6	SF_6	—	Extremely F-rich	Very low	Si	Fast Si etch, MEMS/TSV DRIE
NF_3	NF_3	—	Strong F radical source	None	Residual Si/SiN/SiO_2 films	Primarily chamber cleaning; some plasma etch applications
F_2	F_2	—	Direct fluorine source	None	Si-containing materials	Specialized etch/cleaning applications

Practical Application of Each Important Etching gas

Gas	Why it is selected	Typical process	Key advantage	Main limitation
CF_4	Generates abundant F radicals	SiO_2/SiN/Si etch	High etch rate, simple chemistry	Relatively weak sidewall passivation
CHF_3	H scavenges F and promotes CF_x-rich chemistry	Oxide/nitride RIE	Better selectivity and anisotropy than CF_4	More polymer/residue
C_4F_8	Excellent fluorocarbon polymer precursor	HAR oxide; Bosch DRIE	Excellent sidewall passivation	Polymer buildup; needs careful balance
C_4F_6	Very carbon-rich unsaturated fluorocarbon	Advanced oxide / low-k etch	Excellent profile and selectivity control	Strong polymerization; complex process window
SF_6	Generates large amount of reactive F	Si etch, MEMS DRIE, TSV	Extremely high Si etch rate	Without passivation can be isotropic
NF_3	Efficient source of reactive fluorine	Chamber clean, remote plasma clean	Efficient removal of Si-containing deposits	Generally not the primary choice for profile-controlled fluorocarbon etching

Gas Selection for Etching by Semiconductor application

Process / material	Common gases	Purpose / reason
Si etching	SF_6, CF_4, Cl_2/HBr	SF_6 provides very high F radical concentration
MEMS Si DRIE – Bosch	SF_6 + C_4F_8	SF_6 = Si etch; C_4F_8 = sidewall passivation
TSV Si DRIE	SF_6 + C_4F_8	Deep, highly anisotropic Si etching
SiO_2 etching	CF_4, CHF_3, C_4F_8	Balance F etching with CF_x passivation
SiN_x etching	CF_4, CHF_3, CH_2F_2 + additives	Selectivity/profile control
Contact-hole oxide etch	CHF_3, C_4F_8, C_4F_6 + Ar/O_2/CO	High anisotropy and Si/PR selectivity
HAR oxide etch	C_4F_6, C_4F_8 mixtures	Strong sidewall protection and profile control
Low-k dielectric etch	CF_4, C_4F_8, C_4F_6 mixtures	Control profile while limiting dielectric damage
Chamber cleaning	NF_3, sometimes CF_4/C_2F_6	Generates F radicals to remove deposited Si-containing films

16. Chlorine/Bromine-Based Etching (Cl_2/BCl_3)

These are extremely important for **metal and III-V etching**.
For example: **Al + Cl → $AlCl_x$ → volatile product**
Typical recipe concept: **Cl_2 + BCl_3 + Ar**
where:
Cl_2 → primary chemical etchant
BCl_3 → chlorine chemistry + oxygen/moisture scavenging and surface chemistry
Ar → physical ion bombardment

HBr is especially important for **Si/poly-Si gate etching**.
It provides strong **sidewall passivation and profile control**, enabling highly anisotropic structures.
A classic poly-Si etch chemistry is: **HBr + Cl_2 + O_2**
where:
Cl_2 → etching
HBr → anisotropy/profile/CD control
O_2 → passivation/polymer balance

Chlorine/Bromine-Based Etching *(Application Summary)*

- **Cl-based** chemistry reacts with the wafer material to produce chloride-containing reaction products.
- **BCl_3** chemistry helps remove/scavenge oxygen-containing surface species, allowing the chlorine chemistry to etch the underlying material more effectively.
- **HBr** is particularly important because of its excellent anisotropy/profile and CD control for Si etching.

Etching Gas	Main Reactive Species	Major Materials	Typical Application	Key Advantage
Cl_2	Cl·, Cl^+/Cl-containing ions	Si, poly-Si, Al, Ti, TiN, GaN, AlGaN, GaAs	Gate, metal, III-V etch	High chemical reactivity
BCl_3	Cl·, BCl_x species	Al, Ti/TiN, GaN, AlGaN, III-V	Metal & compound semiconductor etch	Oxygen scavenging + profile control
HCl	Cl·	Si, SiGe, III-V	Selective/surface etching; more common in thermal/epi cleaning than RIE	Selectivity/surface chemistry
HBr	Br·, Br-containing ions	⭐ Si, poly-Si	Gate/fin/Si pattern etching	Excellent anisotropy & CD control
Cl_2 + BCl_3	Cl chemistry	⭐ Al, TiN, GaN	Metal/III-V patterning	Rate + profile control
HBr + Cl_2	Br + Cl	⭐ Poly-Si / Si	Precision Si etching	Rate + anisotropy
HBr + Cl_2 + O_2	Br + Cl + O	⭐ Advanced Si patterning	Gate/fin structures	Sidewall passivation
Cl_2/BCl_3 + Ar	Cl + Ar^+	Metal, GaN/AlGaN	ICP/RIE	Chemical + physical etching

Why Chlorine/Bromine Instead of Fluorine?

F → faster chemical etching
Cl → balanced chemical etching / metal & III-V applications
Br → slower chemistry but excellent anisotropy and CD control

Examples	SF_6, CF_4, NF_3	Cl_2, BCl_3	HBr
Reactivity	⭐⭐⭐ Very high	⭐⭐ High	⭐ Lower
Si etch rate	Very high	High/moderate	Lower
Spontaneous chemical etch	High	Moderate	Lower
Isotropic tendency	Higher	Medium	Lower
Ion assistance importance	Medium	High	High
Sidewall control	Moderate	Good	⭐⭐⭐ Excellent
CD control	Moderate	Good	⭐⭐⭐ Excellent
Metal etching	Limited/material-specific	⭐⭐⭐ Excellent	Limited
III-V etching	Some applications	⭐⭐⭐ Important	Some applications
Typical signature application	Fast Si / dielectric	Metal / III-V / Si	Precision Si/poly-Si

17. O_2 Plasma — Organic Material Removal

Plasma generates: **O_2 → $O\cdot$ + O^+ + other excited species**
These react with carbon-containing materials to form volatile species such as: **CO / CO_2 / H_2O**
O_2 plasma is used for: **Photoresist strip · Ashing · Descum · Organic residue removal · Polymer removal**

Application	Target Material / Contamination	Role of O_2 Plasma	Typical Process / Purpose	Key Consideration
Photoresist stripping / ashing	Photoresist	⭐ Oxidizes organic PR	Remove PR after etch or implantation	Most common application
Descum	Thin PR residue	Removes residual polymer	After lithography/develop	Mild plasma preferred to protect substrate
Post-etch polymer removal	Fluorocarbon CF_x polymer	Oxidizes carbon-rich residue	After oxide/dielectric RIE	May require mixed chemistry
Organic contamination removal	Hydrocarbons / organic residue	Converts carbon to volatile products	Pre-clean / chamber clean	Surface damage must be controlled
Hard-mask removal	Amorphous carbon / SOC	Carbon oxidation	Pattern-transfer processes	High removal rate possible
BARC removal	Organic bottom anti-reflective coating	Oxidizes organic BARC	Lithography/etch integration	CD control important
Polymer etching	Polyimide, polymers	Reactive organic etch	MEMS / packaging	Rate depends strongly on polymer
Polyimide etch/descum	PI	Oxidative removal	Advanced packaging / RDL	Sidewall/profile control
Dry-film resist removal	Organic resist	Oxidation	Packaging / PCB / RDL	Thickness can require long ash
Surface activation	Polymer surface	Creates polar functional groups	Adhesion improvement	Often intentionally modifies only surface
Surface cleaning	Organic monolayers/residue	Removes carbon contamination	Before deposition/bonding	Avoid overexposure
Wafer bonding preparation	Organic contamination	Cleans/activates surface	MEMS / wafer bonding	Surface chemistry is critical
Chamber cleaning	Organic chamber deposits	Oxidizes carbon deposits	RIE/PECVD chamber PM	Depends on deposited material

Descum: remove only residual PR at the bottom of developed features.
Ashing: remove essentially the whole photoresist layer.
O_2 RIE: intentionally pattern/etch an organic layer with directional ion assistance.

Additive Gases Used with O_2

Gas Chemistry	Main Function	Typical Application
O_2	Organic oxidation	PR ash/strip
$O_2 + N_2$	Modify plasma/surface chemistry	PR strip / residue removal
$O_2 + H_2$	Adjust reducing/oxidizing chemistry	Specialized residue clean
$O_2 + CF_4$	Adds F radicals	Difficult polymer/residue removal
$O_2 + SF_6$	Adds strong F chemistry	Specialized residue/material removal
$O_2 + Ar$	Adds physical bombardment	Polymer etch / surface clean
$O_2 + He$	Plasma/dilution control	Ashing applications
H_2/N_2	Reducing alternative to O_2	When oxidation of exposed materials is undesirable

O_2 → attacks carbon/organic component, While **CF_4 → provides F radicals**
The combined plasma can attack more complex **C/F/Si-containing residues** than O_2 alone. However, adding fluorine changes the selectivity and can attack exposed Si-containing materials, so it must be carefully controlled.

After certain metal etches or in advanced interconnect structures, a reducing or oxygen-free plasma clean may be preferred instead of conventional O_2 ashing to prevent Cu/Al/Ti oxidation (and plasma damage)

18. Argon (Ar) — Physical Etching Component

Ar plasma → Ar^+ → physical sputtering

When Ar enters the plasma: **Ar + e^- → Ar^+ + $2e^-$**

The positively charged Ar^+ ions are accelerated across the plasma sheath toward the wafer.

Ar^+ ↓ ↓ ↓ → wafer

The ions transfer momentum to surface atoms:

Ar^+ + Surface → momentum transfer → surface atom/product ejection

This is called **sputtering** or **physical etching**.

Unlike F, Cl or Br: **F· + Si → SiF_4↑** = chemical etching

whereas: **Ar^+ + Si → Si atom ejected** = physical sputtering

Chemical reaction + Ar^+ bombardment → ion-assisted chemical etching

Reactive gas + Ar

This is usually more powerful:

Chemical reaction + Ar^+ bombardment → ion-assisted chemical etching

For example: **Cl_2 + Ar**

Cl radicals chemically react with the material while Ar^+ helps break bonds and remove products.

Common Ar Gas Mixtures and Applications

Plasma Chemistry	Ar Function	Typical Application
Cl_2 + Ar	Ion bombardment	Si, metal, III-V etch
Cl_2 + BCl_3 + Ar	⭐ Physical assistance	GaN / AlGaN / metal etch
HBr + Cl_2 + Ar	Profile control	Si/poly-Si
CF_4 + Ar	Physical assistance	Si/SiO_2/SiN
CHF_3 + Ar	Polymer removal at feature bottom	Dielectric etch
SF_6 + Ar	Enhances physical component	Si/SiC
O_2 + Ar	Physical + oxidative removal	PR/polymer/organic
Ar only	⭐ Pure sputtering	Pre-clean / ion milling

Why Ar Is Important for Low-Volatility Etch Products
Suppose: **Material + Cl → low-volatility MCl_x**
If MCl_x does not readily desorb: **MCl_x remains → blocks surface → etch rate decreases**
Adding Ar provides: **Ar^+ → MCl_x surface → sputtering/desorption ↑**
Therefore: Ar helps remove reaction products that are not sufficiently volatile by chemical processes alone.

Chemical vs Physical vs Ion-Assisted Etching

- F / Cl / Br / O → CHEMISTRY
- Ar^+ → PHYSICAL BOMBARDMENT
- Reactive Gas = Chemical Reaction + Ar^+ = Physical Bombardment → Ion-Assisted Etching → Better Anisotropy & Difficult-Material Removal

Etch Mode	Example	Main Mechanism	Selectivity	Directionality
Chemical	SF_6 → Si	F radical reaction	⭐⭐⭐	⭐
Physical	Ar	Ar^+ sputtering	⭐	⭐⭐⭐
Ion-assisted chemical	Cl_2 + Ar	Chemistry + bombardment	⭐⭐	⭐⭐⭐
Polymer-assisted RIE	CHF_3/C_4F_8 + Ar/ions	Etch + passivation + bombardment	⭐⭐⭐	⭐⭐

Gas Used in Conventional Packaging & Advanced Packaging

Gases Used in Conventional & Advanced Semiconductor Packaging

Conventional Packaging: N_2 + CDA + Forming Gas + Ar/O_2 + He
Advanced Packaging: all of the above + Ar + H_2 + SF_6/C_4F_8 + fluorocarbon/halogen gases + CVD/ALD precursor gases

The important trend is that **advanced packaging is becoming increasingly similar to wafer fabrication**: once the flow includes **RDL, UBM, TSV, Cu pillar, wafer-level packaging, and hybrid bonding**, the facility needs increasingly sophisticated **UHP gas distribution, specialty-gas cabinets, VMB/VMP, gas detection, and point-of-use abatement**, rather than only the relatively simple N_2/CDA infrastructure of traditional assembly.

Group	Main Gases	Main Packaging Function
① Inert / Protective	N_2, Ar	Prevent oxidation, purge, thermal processing
② Reducing	N_2/H_2, Ar/H_2, H_2	Reduce Cu/solder/metal oxides
③ Plasma Clean / Activation	O_2, Ar, Ar/O_2, CF_4/O_2	Remove organics and activate bonding surfaces
④ TSV / Wafer-Level Etch	SF_6, C_4F_8	Deep Si etching for TSV
⑤ Test / Specialty	He, CO_2, specialty gases	Hermetic leak test, laser and specialized processes

Gas usage by packaging process

Packaging Process	Main Gas(es)	Main Purpose
Wafer thinning / backgrind	N_2, CDA	Drying, equipment/pneumatic support
Wafer cleaning	N_2, CDA, sometimes CO_2-related cleaning	Drying / contamination & charge control
Plasma cleaning	Ar, O_2, Ar/O_2	Remove organics and activate surfaces
Die attach	N_2, forming gas	Prevent oxidation during heating
Wire bonding	N_2, forming gas, Ar/O_2 plasma	Pad cleaning / oxidation control
Cu wire bonding	N_2-based shielding gas	Protect heated Cu ball/surface from oxidation
Solder reflow	N_2	Reduce oxidation and improve solder wetting
Fluxless/reducing reflow	H_2-containing atmosphere	Reduce metal oxide
Molding / curing	N_2 / CDA	Purge / process atmosphere / pneumatics
Hermetic package	N_2, Ar, He	Package atmosphere / leak testing
RDL lithography	N_2, CDA, O_2/Ar	Coat/develop support, descum
UBM PVD	Ar, N_2	Ti/TiN/Ta etc. sputtering
Cu seed PVD	Ar	Cu seed deposition
Cu electroplating	Usually no main process gas	Cu deposition from liquid electrolyte
Cu pillar plating	Usually no main process gas	Cu/Ni/SnAg electroplating
RDL / UBM etching	Cl_2/BCl_3 or other chemistry depending on film	Metal patterning
TSV DRIE	SF_6 + C_4F_8	Deep Si via formation
TSV liner deposition	Process-dependent CVD/ALD gases	SiO_2/barrier/liner deposition
Hybrid bonding pre-clean	Ar, O_2, Ar/O_2, specialized plasma	Surface cleaning & activation
Cu-Cu bonding	N_2, Ar, H_2-containing ambient depending on process	Prevent Cu oxidation / promote clean interface
Laser drilling	N_2/CDA; laser-specific gas depending on system	Assist gas / debris removal
Package leak test	He	Helium fine-leak detection

Gas Used in Metrology & Inspection

Metrology & Inspection Gases — Functional Groups, Applications & Tools

N_2 / CDA → Support the metrology tool
Ar / Ar^+ → Analyze surfaces & depth profiles
Xe / Ga / He / Ne → Image, mill & cross-section

O_2 / Ar-O_2 → Clean samples & chambers
He / Calibration gases → Leak test & calibrate

Functional Group	Main Gases	Main Function	Metrology / Inspection Application	Typical Tools / Equipment
① Utility / Purge	N_2, CDA	Purge, moisture/contamination control, pneumatics, wafer handling	Dimensional metrology, optical inspection, defect inspection, film measurement	CD-SEM, Optical Inspection, Defect Inspection, Overlay Metrology, Ellipsometer, Reflectometer, XRR/XRD, AFM, Profilometer
② Surface Analysis / Depth Profiling	Ar / Ar^+	Ion sputtering, surface cleaning, layer-by-layer material removal	Elemental composition, chemical-state analysis, contamination analysis, depth profiling	XPS, AES, SIMS, ion sputter/depth-profile systems
③ FIB / Ion Microscopy	Xe, He, Ne; Ga as LMIS*	Ion-beam imaging, milling, cross-sectioning and sample preparation	Failure analysis, cross-section inspection, defect localization, TEM lamella preparation, nanoscale imaging	Ga-FIB, FIB-SEM, Xe-PFIB, Helium Ion Microscope (HIM), Ne ion microscope/FIB
④ Chamber / Sample Cleaning	O_2, Ar, Ar/O_2	Remove hydrocarbon/organic contamination and clean surfaces	Pre-analysis cleaning, chamber cleaning, sample-holder cleaning, contamination reduction	SEM, TEM, FIB-SEM, XPS, AES, plasma cleaner
⑤ Calibration / Specialized Analysis	He, certified calibration gas mixtures, instrument-specific gases	Leak detection, instrument calibration, detector verification, environmental/AMC monitoring	Vacuum leak testing, AMC measurement, gas analyzer calibration, specialized analytical measurement	He Leak Detector, AMC Monitor, FTIR Gas Analyzer, RGA, GC/MS, gas detector

* Ga is not a gas; conventional Ga-FIB uses a liquid metal ion source (LMIS).

Gas Storage & Piping Materials

Bulk Gas Storage in a Wafer Fab

Bulk gases are generally specified by liquid storage volume or Nm^3 capacity, while specialty gases are commonly specified by cylinder water capacity and net product weight.

Compressed gas: Cylinder volume → pressure determines available Nm^3
Liquefied gas: Cylinder volume → liquid density + allowable filling ratio determine **kg of product**

N_2	Cryogenic liquid	LIN tank + vaporizer or on-site ASU	~10–100+ m^3 liquid
O_2	Cryogenic liquid	LOX tank + vaporizer	~5–50+ m^3 liquid
Ar	Cryogenic liquid	LAr tank + vaporizer	~3–30+ m^3 liquid
H_2	Compressed gas / liquid / on-site generation	Tube trailer, MEGC, bulk storage	Site dependent
He	Compressed gas / liquid	Cylinder bundle, tube trailer, liquid container	Site dependent
CO_2	Liquefied	Bulk CO_2 tank	Several–tens of tonnes
CDA	Generated on site	Compressor + dryer + receiver	Not normally delivered in cylinders

Specialty Gas Cylinders

For specialty gases, **kg of product** is often more useful than Nm³ because many semiconductor gases are supplied as **liquefied gases**.

Container	Approx. Water Volume	Typical Product	Representative Fill Quantity	Typical Use
Lecture bottle / small cylinder	1–5 L	Calibration mixtures, specialty gases	<1–several kg	R&D / calibration
10–20 L cylinder	10–20 L	PH_3 mixtures, B_2H_6 mixtures, specialty gases	Gas-dependent	Low consumption
40–50 L cylinder	~40–50 L	SiH_4, NH_3, HCl, Cl_2, WF_6, NF_3, CF_4, SF_6	~5–50+ kg depending on gas	Standard fab specialty gas
Y-cylinder	~400–500 L	NH_3, NF_3, N_2O, CF_4, SF_6, SiH_4 and other high-use gases	~100–500+ kg, highly gas-dependent	Centralized BSGS
Ton container	Large liquefied-gas vessel	Certain high-volume liquefied gases	Around hundreds of kg to ~1 tonne class	High-volume centralized supply
ISO container	Gas-specific	High-volume specialty/bulk-specialty gas	Several tonnes	Very high-volume fabs

Summary of Gas Storage Container

Specialty Gas — Small	Small cylinder	1–10 L WC	Calibration / dopant mixtures
Specialty Gas — Standard	40–50 L cylinder	~7–10 Nm^3 compressed gas	SiH_4, Cl_2, HCl, specialty mixtures
Bulk Specialty Gas (BSG)	Y-cylinder	~400–500 L WC	NF_3, NH_3, N_2O, SiH_4, CF_4
High-volume Gas	Bundle / MEGC	Hundreds–thousands Nm^3	H_2, He, N_2
Bulk Compressed Gas	Tube trailer	Thousands Nm^3	H_2, He
Bulk Cryogenic Gas	Cryogenic tank	10–100+ m^3 liquid	N_2, O_2, Ar
On-site Generated	ASU / PSA / compressor	Continuous	N_2, O_2, CDA

Main Gas Piping Materials

Gas piping is one of the most critical parts of semiconductor gas delivery, because the piping material, surface finish, joining method, cleanliness, and compatibility directly affect **gas purity, particle generation, moisture, corrosion, leakage, and fab safety**.

Material	Characteristics	Typical Use	Advantages	Limitations
SS304L	Austenitic stainless steel	Some bulk/utility gases	Lower cost	Less corrosion resistant than 316L
SS316L	Low-carbon Mo-containing stainless steel	Most semiconductor process gases	Excellent corrosion resistance, weldability, cleanliness	Higher cost
316L BA	Bright Annealed 316L	Bulk/UHP gases	Smooth, clean internal surface	Less optimized than EP for highest-purity applications
316L EP	Electropolished 316L	UHP specialty gases	Very smooth surface, low particles, low adsorption/outgassing	Higher cost
Ni alloys	Nickel-based corrosion-resistant alloys	Aggressive/corrosive/reactive gases	Excellent chemical resistance	Expensive
PTFE / PFA	Fluoropolymer	Selected chemical/vapor/exhaust services	Excellent chemical resistance	Generally not the default for pressurized UHP process-gas distribution
Aluminum	Lightweight metal	CDA / selected utility systems	Light, easy installation	Not suitable for many reactive specialty gases
Copper	Good thermal conductivity	Some legacy/general gas services	Easy fabrication	Generally avoided for modern UHP semiconductor specialty-gas distribution

Gas Piping Materials Selection by Gas Hazard

Semiconductor gas piping = Material Compatibility + UHP Surface + Clean Welding + Leak Tightness + Contamination Control + Safety

316L EP Stainless Steel = the workhorse material for semiconductor UHP specialty-gas delivery. However, **316L is not universally compatible with every gas under every pressure, moisture level, temperature, and concentration**.

Gas Category	Examples	Preferred Approach
Inert	N_2, Ar, He, Ne	316L BA/EP
Oxidizer	O_2	316L + oxygen cleaning
Flammable	H_2, SiH_4	316L EP + leak-tight welded system
Toxic	PH_3, AsH_3	316L EP + containment/GDS
Corrosive	HCl, Cl_2, HBr	316L EP or compatible higher alloy
Fluorinating / highly reactive	F_2, ClF_3	Gas-specific material compatibility is critical
Metal-organic precursor	TMA, TMGa, TMIn, TEMAH	316L EP, often heated delivery
Condensable vapor	TEOS and other liquid precursors	316L EP + temperature-controlled line

BA: Bright Annealed
EP: Electropolished

Bright annealed vs. Electropolished SS Tubing

Rough surface→ larger effective surface area→ more H_2O adsorption→ more particles→ more corrosion sites→ slower purge-down→ higher contamination risk

Industrial SS tubing→ rougher surface→ more adsorption/particle sites

Bright annealing is a heat-treatment process performed under a controlled atmosphere to prevent heavy oxidation of the stainless-steel surface (**smoother)**.

Electropolishing is an electrochemical process that preferentially removes microscopic surface peaks from stainless steel (**very smooth with Ra =0.13-0.25 μm + low contamination)**.

Item	BA	EP
Full name	Bright Annealed	Electropolished
Basic process	Heat treatment in a controlled/reducing atmosphere	Electrochemical removal of a thin surface layer
Main purpose	Produce a clean, smooth, oxide-controlled surface	Produce an ultra-smooth, clean, passive surface
Surface roughness	Low	Very low
Particle generation	Low	Very low
Moisture adsorption	Low	Lower
Outgassing	Low	Very low
Corrosion resistance	Good–excellent	Excellent
Purge / dry-down	Good	Faster / better
Cost	Lower	Higher
Typical fab use	Bulk/UHP utility gas	Critical UHP & specialty gas
Typical material	304L / 316L	Usually 316L

Smoother EP surface → faster dry-down + lower moisture + lower particles + improved corrosion resistance

Joining methods for UHP Piping

For critical UHP process gas: **Minimize fittings → maximize orbital welds**.
Every mechanical joint adds potential: **Leak + particle + dead volume + contamination risk**
For hazardous gases, the design may additionally require:
Double containment + continuous exhaust + gas detection + automatic shutoff

Joining Method	Typical Application	Semiconductor Suitability
Orbital GTAW welding	Main UHP gas distribution	★★★★★
VCR metal gasket fitting	Equipment/component connections	★★★★★
Face-seal fitting	Gas cabinet, VMB/VMP, tool connection	★★★★★
Compression fitting	Utility/noncritical service	★★★
Threaded fitting	General industrial service	★
Brazing	Certain utility systems	★★

GTAW: Gas Tungsten Arc Welding

Gas Flow Units Used in Semiconductor Manufacturing

Unit	Full Name	Meaning / Basis	Typical Flow Range	Typical Semiconductor Application
sccm	Standard Cubic Centimeter per Minute	Gas volume referenced to specified standard T/P conditions	~0.1–10,000 sccm	Most common process-tool gas unit; CVD, ALD, etch, implant, RTA
slm / SLM	Standard Liter per Minute	Standard liters of gas per minute	~0.1–1,000+ slm	High-flow process gases, purge, chamber clean
Nm^3/h	Normal Cubic Meter per Hour	Volume at defined "normal" reference conditions	~1–10,000+ Nm^3/h	Fab bulk-gas distribution, N_2, O_2, Ar, H_2
Nm^3/min	Normal Cubic Meter per Minute	Normalized volumetric flow per minute	High flow	Large bulk-gas systems
Sm^3/h	Standard Cubic Meter per Hour	Volume referenced to specified "standard" conditions	High flow	Bulk gas / facility engineering
SCFH	Standard Cubic Feet per Hour	Standard ft^3 per hour	Application dependent	Common in U.S.-based gas/facility specifications
SCFM	Standard Cubic Feet per Minute	Standard ft^3 per minute	High flow	CDA, N_2, exhaust and facility systems
L/min	Liter per Minute	Actual volumetric flow unless reference conditions are defined	Application dependent	Utility/process engineering
m^3/h	Cubic Meter per Hour	Actual volumetric flow	Medium–high	Utility piping, exhaust, ventilation
kg/h	Kilogram per Hour	Mass flow rate	Application dependent	Bulk gas/liquid supply, precursor consumption
kg/day	Kilogram per Day	Daily mass consumption	Application dependent	Specialty-gas consumption / fab planning
kg/s	Kilogram per Second	SI mass flow	Engineering calculations	Large-system/process modeling

sccm is NOT simply cm^3/min, ccm = actual cubic centimeters per minute under actual conditions.
sccm = cubic centimeters per minute referenced to specified **standard temperature and pressure**.

Most Important Units to Remember

Unit	Scale	Where You See It
sccm	Very small → medium	Process chamber / MFC
slm	Medium	Process tool / purge / chamber clean
Nm^3/h	Large	Bulk gas / fab facility
SCFM	Large	CDA / facility / exhaust
kg/h or kg/day	Consumption	Specialty gas / precursor / supply planning

Gas Hazardous & Treatment

High-Hazard Gases Used in Semiconductor Manufacturing

Toxic	AsH_3, PH_3, B_2H_6, HF, F_2, Cl_2	Poisoning / severe exposure	GDS + Gas Cabinet + Abatement
Pyrophoric	SiH_4, B_2H_6, GeH_4	Spontaneous ignition	Flow limitation + isolation + abatement
Flammable	H_2, SiH_4, B_2H_6, PH_3	Fire / explosion	Ventilation + detection + interlock
Corrosive	HF, HCl, HBr, Cl_2, F_2, WF_6	Chemical burns / equipment corrosion	Compatible UHP piping + scrubber
Oxidizing	O_2, O_3, F_2, ClF_3, NF_3, N_2O	Accelerated combustion / violent reaction	Material compatibility + segregation
Asphyxiant	N_2, Ar, He, CO_2, SF_6	O_2 displacement	Ventilation + O_2 deficiency monitor

A single gas can have **multiple hazards simultaneously**:
B_2H_6 = Toxic + Pyrophoric + Flammable
F_2 = Toxic + Corrosive + Strong Oxidizer
ClF_3 = Toxic + Corrosive + Extremely Reactive Oxidizer
SiH_4 = Pyrophoric + Flammable
PH_3 = Extremely Toxic + Flammable
N_2 = Non-toxic + Non-flammable, but Asphyxiant

This is why semiconductor gas safety should be based on a **multi-hazard assessment rather than simply labeling a gas "dangerous" or "safe."**

Exhaust segregation is critical

A fab should not combine all exhaust streams into one duct. Improper mixing can create fire, precipitation, corrosion, toxic reaction products, or duct blockage.

Exhaust System	Typical Sources	Why Separate?
Acid Exhaust	HF, HCl, HBr, HNO_3	Corrosion / acid treatment
Alkaline Exhaust	NH_3, NH_4OH	Avoid unwanted acid-base reaction/deposition
Solvent/VOC Exhaust	IPA, acetone, resist solvents	Fire/VOC control
Toxic Exhaust	PH_3, AsH_3, toxic process gases	High hazard
Pyrophoric/Flammable Exhaust	SiH_4, H_2-related streams	Fire/explosion
PFC Process Exhaust	CF_4, C_4F_8, SF_6, NF_3	Requires destruction technology
General Exhaust	Heat, nonhazardous emissions	No special chemical treatment

The 4 Major Scrubber Types

Type	Principle	Best Suited For	Major Advantage	Main Limitation
💧 1. Wet	Absorb + neutralize using water/chemicals	HF, HCl, HBr, Cl_2, NH_3	Simple, effective for soluble gases	Generates wastewater; cannot efficiently destroy many stable PFCs
🔥 💧 2. Burn-Wet	Combust/oxidize → wet scrub	SiH_4, B_2H_6, H_2, hydrides	Excellent for combustible/pyrophoric gases	Fuel/energy consumption; produces particles/wastewater
⚡ 💧 3. Plasma-Wet	Plasma decomposition → wet scrub	CF_4, C_4F_8, NF_3, SF_6	Effective for difficult fluorinated gases / GHG control	Higher energy and equipment cost
🧱 4. Dry	Adsorption / chemisorption on solid media	PH_3, AsH_3, halogens, selected toxic gases	No scrubber wastewater; compact	Spent media requires replacement and hazardous-waste management

Summary — POU Abatement, Scrubbers & Exhaust

Point-of-use (POU Abatement = treat concentrated process gas (Destroy/Convert) at the tool
Exhaust = safely contains and transport the remaining gases
Scrubber = remove/destroy contaminants (Capture/Neutralize) before release

POU Abatement	Point-of-Use Abatement installed close to an individual process tool	Destroy, convert or capture hazardous gases before they enter the facility exhaust	SiH_4, NF_3, CF_4, C_4F_8, SF_6, PH_3, AsH_3, NH_3, WF_6, process by-products	CVD, PECVD, ALD, etch, epi, implant, MOCVD	Tool / subfab
Wet Scrubber	Uses water or chemical solution to absorb and neutralize contaminants	Absorption + neutralization + some particle removal	HF, HCl, HBr, Cl_2, NH_3, soluble F/Cl species	Etch, wet bench, CVD, acid/base exhaust	POU or central facility
Burn-Wet Scrubber	Combustion/oxidation stage followed by wet scrubbing	Burn/oxidize → capture/neutralize	SiH_4, B_2H_6, H_2, hydrides and combustible process gases	CVD, PECVD, epi, some dopant processes	Mainly POU / subfab
Plasma-Wet Scrubber	High-energy plasma decomposes stable gases, followed by wet scrubbing	Destroy → capture → neutralize	CF_4, C_2F_6, C_4F_8, NF_3, SF_6 and fluorinated by-products	Plasma etch, CVD chamber clean	Mainly POU / subfab
Dry Scrubber / Adsorber	Reactive solid media chemically adsorbs or converts hazardous gases	Chemisorb / capture without liquid scrubbing	PH_3, AsH_3, H_2S, halogens and other compatible toxic gases	Implant, specialty gas systems, selected CVD/epi	POU / gas room / facility
Exhaust System	Duct/piping and fans that safely transport process emissions	Contain + transport + maintain negative pressure	Acid, alkaline, solvent, toxic, process and general exhaust	Entire cleanroom/fab	Tool → subfab → facility → stack

Summary

1. SOURCE & SUPPLY



- Air Separation Unit (ASU)
- On-site generation
- Specialty gas suppliers

2. STORAGE



- Cylinders, Y-cylinders, Bundles, Bulk Tanks
- Gas Cabinet / Gas Rack
- Safe & compliant storage

3. DELIVERY



- VMB / VMP / Gas Piping
- Clean materials (BA/EP)
- Leak-tight & high purity

4. PROCESS APPLICATION



- Deposition
- Etching
- Doping
- Thermal / Anneal
- Lithography
- CMP, Cleaning
- Packaging, Metrology

5. EXHAUST & ABATEMENT



- POU Abatement
- Scrubbers (Wet, Burn-Wet, Plasma, Dry/Adsorber)
- Safe discharge

THE GOAL

- Safety for People
- Uptime for Tools
- Yield for Products
- Sustainability for Future

TYPES OF GASES

Bulk Gases (High Volume)

N_2 O_2 H_2 Ar He

CDA, CO_2, Ne, Kr, Xe

Mainly for environment control, process support, purging, cooling and protection.

Specialty Gases (Low Volume, High Purity)

SiH_4 NH_3 WF_6 HCl PH_3 B_2H_6

Precursors, reactants, etchants, dopants, chamber cleaning and specialty applications.

GASES BY FUNCTION (Examples)

Carrier / Inert	N_2, Ar, He
Precursor	SiH_4, TEOS, NH_3, WF_6, TMA
Reactant	O_2, H_2, NH_3, HCl
Etchant	CF_4, SF_6, Cl_2, BCl_3, NF_3
Dopant	PH_3, AsH_3, B_2H_6
Purging / Dilution	N_2, Ar
Chamber Clean	NF_3, O_2 (ash), ClF_3

GAS SAFETY – KNOW THE HAZARDS

Toxic	HCl, Cl_2, NF_3, PH_3, AsH_3, B_2H_6, WF_6, NH_3
Pyrophoric	PH_3, B_2H_6, SiH_4, F_2
Flammable	H_2, SiH_4, NH_3, C_2H_2
Corrosive	HCl, HF, Cl_2, BCl_3, NH_3
Oxidizing	N_2O, O_2, ClF_3, F_2
Asphyxiant	N_2, Ar, He, CO_2

SAFETY ESSENTIALS

- Gas Detection System (GDS)
- Interlocks & Ventilation
- Emergency Shutdown (ESD)
- Proper Materials & Piping
- Training & Standard Operation

MAJOR APPLICATIONS ACROSS SEMICONDUCTOR PROCESSES

Cleaning	Deposition	Etching	Doping	Metallization	Thermal / Anneal	Lithography	CMP	Packaging	Metrology & Inspection
O_2 plasma, H_2O_2, NH_3	SiH_4, TEOS, NH_3, TMA, WF_6	CF_4, SF_6, Cl_2, NF_3	PH_3, AsH_3, B_2H_6	NH_3, N_2, Forming Gas	N_2, Ar, H_2, Forming Gas	Ar (purge), N_2	N_2 (dry), CO_2 (slurry)	N_2, Forming Gas, Plasma Clean	N_2, Clean Air

GAS QUALITY & CONTROL

UHP		Purity Verification	Flow & Pressure	Particle Control
ppt–ppb impurity level	Moisture is enemy of yield & reliability	Regular analysis & monitoring	Stable, accurate & consistent	Clean piping, no particle shed

KEY PRINCIPLES

- Right Gas for the Right Process
- Right Purity, Right Flow, Right Pressure
- Safe Handling, Reliable Delivery
- Effective Abatement, Minimal Impact

SUCCESS FACTORS

System Integration · Material Compatibility · Preventive Maintenance · Data & Monitoring · People & Training

REMEMBER

Gases may be invisible, but they are critical to every step of semiconductor manufacturing.

Purity + Safety + Reliability = High Yield & Sustainable Fab

Related media for semiconductor industry



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